Chapter 1: Biological Molecules

PM&DC Syllabus

- **Biological molecules:** Define and classify biological molecules.
- Discuss the importance of biological molecules
- **Biological Importance of Water:** Describe biologically important properties of water (polarity, hydrolysis, specific heat, water as solvent and reagent, density, cohesion/ionization)
- **Carbohydrates**: Discuss carbohydrates: monosaccharaides (glucose), oligosaccharides (cane sugar, sucrose, lactose), polysaccharides (starches, cellulose, glycogen)
- **Proteins**: Describe proteins: amino acids, structure of proteins
- **Lipids:** Describe lipids: phospholipids, triglycerides, alcohol and esters (acylglycerol)
- **Ribonucleic acid (RNA):** Give an account of structure and function RNA
- **Conjugated molecules:** Discuss conjugated molecules (glycol lipids, glycol proteins)
- **DNA:** Described in more detail chapter 'DNA and Chromosomes'

NUMS Syllabus

a. Introduction to biological molecules
b. Water
c. Carbohydrates
d. Proteins
e. Lipids
f. Conjugated molecules (glycolipids, glycoproteins)

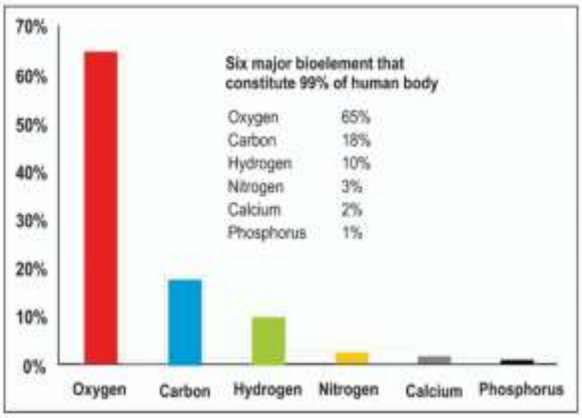
Introduction to Biochemistry

- **Biological molecules** are different chemical compounds of living beings.
- **Biochemistry** is the branch of biology which deals with the study of chemical components and chemical processes in living organisms.

Chemical composition of protoplasm

(BTB)

- Protoplasm is the living content of the cell that is surrounded by a plasma membrane.
- It is a general term for cytoplasm and nucleoplasm.
- Approximately **25 elements** out of 92 naturally occurring elements of earth are found in living beings. These are called **bio elements**. Or biogenic elements.
- Human body is composed of only 16 of these bio elements.
- The **six** commonest bio elements that constitute 99% of protoplasm are called **major** bio elements.



Minor elements include: potassium (0.35%), sulphur (0.25%), chlorine (0.15%), sodium (0.15%) and magnesium (0.05%). **Trace elements** include iron, copper, manganese, zinc

Iron (0.04%), iodine (trace)

Proportion of various bio elements in human body.

- **Minor** bio elements are those that are found as less than 1% whereas those that are found as less than 0.01% of the protoplasm are called **trace elements.**
- They are also called **dietary elements.**
- The **survival** of an organism depends upon its ability to take up some chemicals from its environment and use them to chemical of its living matter.
- The bio elements are combined with each other and can form thousands of different biomolecules which may be:

1. **Inorganic** (water, minerals, carbon dioxide, acids, bases and salts)
2. **Organic** (carbohydrates, lipids, proteins and nucleic acids).

Proportion of various biomolecules in bacterial and mammalian cells

Biomolecules	Bacterial Cell	Mammalian Cell

Water	70%	70%
Protein	15%	18%
Carbohydrates	3%	4%
Lipids	2%	3%
DNA	1%	0.25%
RNA	6%	1.1%
Other organic molecules (enzymes, hormones, metabolites)	2%	2%
Inorganic ions (Na^+, K^+, Ca^{++}, Mg^{++}, Cl^-, SO_4^{2-})	1%	1%

Classification of bioelements (25) → On the basis of presence →
- Major forming 99% of body including C, H, O, N, Ca, P
- Minor forming 1% body e.g., S, Mg, etc.
- Trace / dietary forming about 0.1% e,g., Zn, Mn, etc.

Macro-organic molecules

- **Carbohydrates** are composed of C, H, O and provide fuel for the metabolic activities of the cell. Carbohydrates are present in the cytoplasm of the cells.

(KPK)

- **Proteins** are the **most abundant** organic compounds in protoplasm and body.
- They have both structural and functional role in the body.
- Proteins are present in the membranes, ribosomes, cytoskeleton and enzymes of the cell.
- **Lipids** are **heterogenous** groups of hydrophobic compounds. Lipids are present in the membranes and cytoplasm of the cell
- **Nucleic acids** (DNA and RNA) are **most essential** organic compounds, for living organisms. The nucleic acid DNA is present in the chromosome. The nucleic acid RNA is present in the nucleoplasm and cytoplasm.

1. C-H bond	Hydrocarbon (potential source of chemical energy for cellular activities)
2. C-N bond	Peptide bond (forms proteins which are very important due to their diversity in structure and function)
3. C-O bond	Glyosidic bond (provides stability to complex carbohydrate molecules)

Metabolism
- **Anabolism**
  - Needs energy
  - Simpler compounds join
  - Complex compounds are formed
  - Such as photo-synthesis
- **Catabolism**
  - Energy is released
  - Larger molecules break
  - Smaller simpler molecules formed
  - Such as respiration

Metabolic reactions of cell

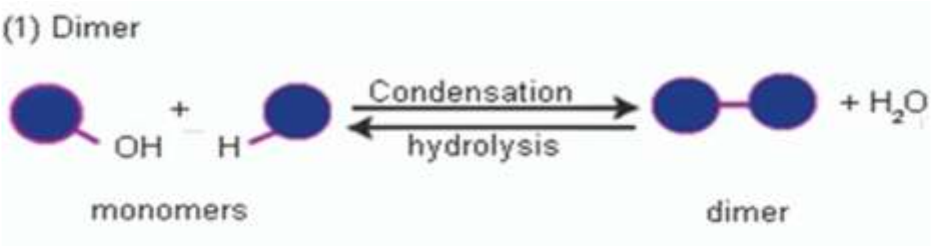
- All the reactions taking place within a cell are collectively referred to as **metabolism.**
- Those reactions in which simpler substances are combined to form complex substances are called **anabolic reactions**. Anabolic reactions need energy.
- Energy is released by the breakdown of complex molecules into simpler ones, such reactions are called **catabolic reactions.**

Note

- Don't confuse involvement of water in hydrolysis with making a solution, in which role of water is to act as a solvent, rather than taking part in chemical reaction.
- Also do not assume that this breakdown releases energy, which is usually produced when the simpler substances are oxidized in respiration.
- Hydration is yet another completely different process, involving the addition of water but not the breaking of bonds.

Condensation and hydrolysis

Condensation	Hydrolysis
• Condensation is the combination of monomers into polymer by the removal of water.	• The hydrolysis is essentially the reverse of condensation i.e., the breakdown of a polymer into its monomers by the addition of water.
• During condensation, a hydroxyl (OH^-) group is removed from one monomer and a hydrogen (-H) is removed from the other to make water and as a result a bond is synthesized between the monomers. Condensation is also called **dehydration synthesis**.	• During hydrolysis, an (OH^-) group from water is attached to one monomer and (H) is attached to the other monomer. All digestion reactions are example of hydrolysis.
• Condensation does not take place unless the proper enzyme is present, and the monomers are in an activated energy-rich form.	• Actually, all digestion reactions are examples of hydrolysis, which are controlled by enzymes such as carbohydrases, proteases, lipases,



Condensation and hydrolysis

Importance of Water

- Water is the **most abundant** compound in all organisms and forms three fourth of the body in humans.
- It varies from **65 to 89 percent** **(**FTB = 70%, BTB & KPK = 70 to 90%**)** in different organisms.
- Human tissues contain about 20 percent water in bone cells and 85 percent (FTB = 85 to 90%) in brain cells.
- Seeds also contain 20% water.
- contains 88% water.
- Cucumber contains 98% water.
- Jelly fish has exceptionally large amount of water i.e. 99% that's why its body shows transparency.
- It is also used as raw material for photosynthesis.

(KPK)

- Water is essential for existence of protoplasm because protoplasm can't survive if its water content is reduced as low as 10%.
- Water dissolves all minerals present in soil which are absorbed by plant roots and transported to other tissues

Solvent properties

- Due to its polarity, water is an excellent solvent for polar substances and hence called **universal solvent.**
- Ionic substances when dissolved in water, dissociate into positive and negative ions.
- Non-ionic substances having charged groups in their molecules are dispersed in water.
- It is because of this property of water that almost **all reactions** in cells occur in aqueous media.
- Nonpolar organic molecules, such as fats, are insoluble in water and help to maintain membranes which make compartments in the cell.

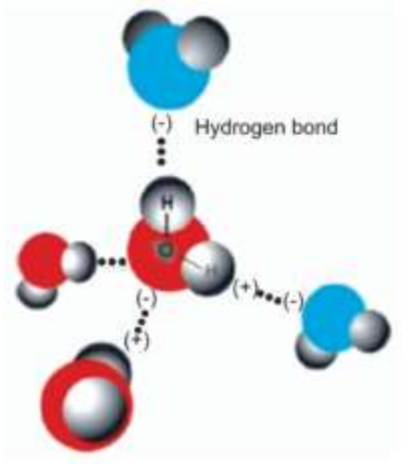
Hydrogen bonding

- The polarity of water molecules makes them interact with each other.
- The charged regions on each molecule are attracted to oppositely charged regions on neighbouring molecules, forming **weak bonds.**
- Since the positively charged region in this special type of bond is always an H atom, the bond is called a **hydrogen bond.**
- This bond is often represented by a dotted line because a hydrogen bond is easily broken.

- The presence of hydrogen bonds among water molecules causes it to remain in liquid state rather than change to ice or steam.

(KPK)

- Without hydrogen bonds, water would boil at **-80** degree Celsius and freeze at **-100** degree Celsius.
- Hydrogen bonds are weaker than covalent bonds but still cause water molecules to remain attached together.



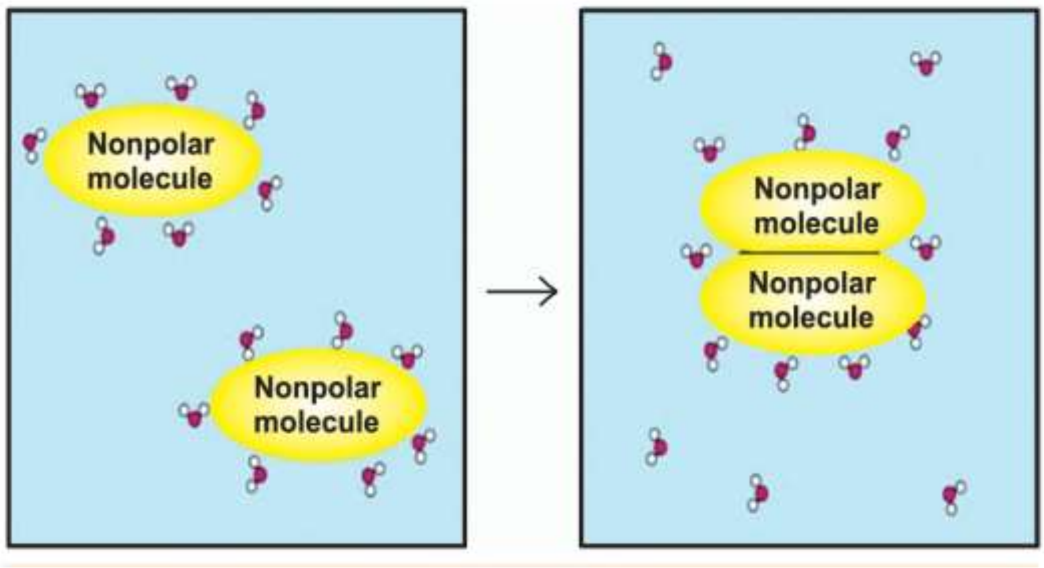
Hydrogen Bonding between water molecules

Cohesion and adhesion

- Cohesion is the intermolecular attraction between **similar** molecules while adhesion is the attraction between **dissimilar** molecules.
- **Cohesion** is the attraction among the water molecules which enables the water molecules to stick together and flow together.
- Water molecules also have attraction to polar surfaces. This attraction is called **adhesion.**
- Both cohesion and adhesion are due to **hydrogen bonds** among water molecules.
- These properties of water enable it to act as **transport medium.**
- Water can fill tubular vessel and still flows so that dissolved molecules are evenly distributed throughout the system

Hydrophobic exclusion

- Hydrophobic exclusion can be defined as reduction of the contact area between water and hydrophobic substances which are placed in water.
- Biologically, hydrophobic exclusion plays key roles in maintaining the **integrity of lipid bilayer membranes.**



Hydrophobic Exclusion

High specific heat capacity

- Water has great ability of absorbing heat with minimum of change in its own temperature due to its **high** specific heat capacity.
- The specific heat capacity of water i.e. the number of calories required to raise the temperature of 1g of water from 15 to 16 degree C is **1.0 calorie (or 4.18 joules)** which is relatively high.
- **Reason** of high specific heat is that much of the energy is used to break hydrogen bonds.
- Water thus works as temperature stabilizer for organisms in the environment and hence protects living material against sudden thermal changes due to this property.
- Due to this property, hot water cools slowly while cool water gets hot slowly.

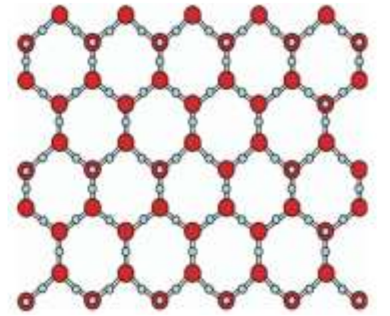
High heat of vapourization

- Water absorbs much heat as it changes from liquid to gas due to its **high** heat of vapourization.

- Heat of vaporization is expressed as calories absorbed per gram vaporized. The specific heat of vaporization of water is **574 kcal/kg.**
- This property plays an important role in the **regulation of heat** produced by oxidation.
- It also provides **cooling effect** to plants when water is transpired, or to animals when water is perspired.
- This is high heat of vapourization of water that gives animals an efficient way to **release excess body heat** in a hot environment.
- Evaporation of only **two ml** out of **one litre** of water lowers the temperature of the remaining **998 ml** by **1** degree C.

Ionization of water

- The dissociation of a molecule into ions is called ionization.
- When water molecule ionizes, it releases an **equal** number of positive hydrogen and negative hydroxyl ions.
- This reaction is reversible but an equilibrium is maintained.
- At **25** degree C the concentration of each of H^+ and OH^- ions in pure water is about **10^{-7} mole/litre.**
- The H^+ and OH^- ions effect and take part in many chemical reactions that occur in the cells.



Lattice like arrangement of ice.

Lower density of ice

- Ice **floats** on water. This is because ice is less dense than water.
- Water is the only substance whose solid form is less dense than liquid.
- The reason of less density is that ice has a **giant structure** and show maximum number of hydrogen bonding among water molecules; hence, they are arranged like a **lattice.**

(KPK)

- During winters ice forms a layer at the top of ponds an lakes.
- **The ice layer** at surface acts as an insulator to prevent water below it from freezing thus preventing living organisms from freezing.
- Water expands at low temperature
- Water has a unique property, as it expands (density ↓↓) when temperature falls below 4 °C.
- Water is most heavy at 4 °C
- Water body freezes on the surface at low temperature.

Protection

- Water is an effective **lubricant** that provides protection against the damage resulting from friction.
- For example, **tears** protect the surface of eye from the rubbing of eyelids.
- Water also forms a **fluid cushion** around organs that helps to protect them from trauma.
- For example, pleural fluid around lungs, pericardial fluid around heart, CSF around brain.

Carbohydrates

- Carbohydrates are the **most abundant** biomolecule in nature.
- The word carbohydrate literally means **hydrated carbons**.
- They are **composed** of carbon, hydrogen and oxygen and the ratio of hydrogen and oxygen is the same as in water i.e. 2:1.
- Their general formula is **$C_x(H_2O)_y$** where x is the whole number from three to many thousands whereas y may be the same or different whole number.
- Chemically, carbohydrates are defined as "**polyhydroxy aldehydes or ketones,** or **complex substances** which on hydrolysis yield polyhydroxy aldehyde or ketone subunits."

(BTB)

- The chemistry of carbohydrate is determined by aldehyde and ketone group e.g. aldehyde is very easily oxidized and hence are **powerful reducing agent.**
- The C-H bonds in the carbohydrate molecules are broken down during respiration and the stored energy in these bonds is released which is made available to the cells for performing various functions.

(KPK)

- The sources of carbohydrates are **green living things** e.g. plants, cyanobacteria, algae and many bacteria.
- These are the primary products of photosynthesis.
- They are stored in cells as reserve food.
- One gram of carbohydrates gives **4.1** kcal of energy.
- Carbohydrates are commonly known as sugar or saccharides.
- Carbohydrates combine with lipids and proteins called glycoproteins and glycolipids.
- Both have structural role in extracellular matrix of animals and bacterial cell wall.
- Both are the conjugated molecule.

Classification of carbohydrates

Carbohydrates

- Monosaccharides: single monomeric sugar
  - Aldoses e.g glucose
  - Ketoses e.g fructose
- Oligosaccharides 2-10 sugar units e.g sucrose
  - Reducing sugar e.g maltose
  - Non-reducing sugars e.g Sucrose
- Polysaccharides containing more than 10 units
  - Homo-polysaccharides having same type of monomeric units. e.g starch , glycogen etc.
  - Hetro-polysaccharides having different type of monomers. e.g agar

Carbohydrates are classified into three groups which are:

Monosaccharides	Oligosaccharides	Polysaccharides
They consist of single saccharide unit.	They are composed of 2 to 10 saccharide units.	They are composed of more than 10 monosaccharide units.
They are simplest carbohydrates; therefore, they cannot be further hydrolyzed.	They have less complex structure, so upon hydrolysis they yield at least 2 and maximum 10 monosaccharides.	They have highly complex structure, so upon hydrolysis they yield at least 11 monosaccharides.
They are highly soluble in water.	They are less soluble in water.	They are generally insoluble in water.
They are sweetest among all carbohydrates.	They are less sweet in taste.	They are tasteless.

Monosaccharides

- Monosaccharides are only **true carbohydrates.**
- All carbon atoms in a monosaccharide except one, have a hydroxyl group.
- The remaining carbon atom is either a part of an aldehyde group or a keto group.
- Their general formula is $C_nH_{2n}O_n$.
- They are white crystalline powders.

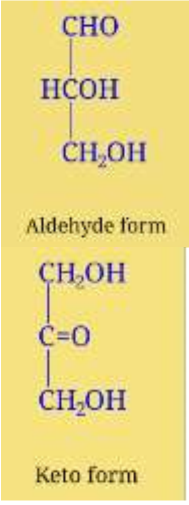
Classification of monosaccharides

- It may be on the basis of functional group or number of carbon atoms.
- **On the basis of functional group,** the monosaccharides containing aldehyde are called **aldoses** while those containing ketone are called **ketoses.**
- The range of **number of carbons** in monosaccharides is **3 to 7**. Hence,monosaccharides are classified into five groups based upon number of carbon atoms i.e., trioses (3C), tetroses (4C), pentoses (5C), hexoses (6C) and heptoses (7C).

(BTB)

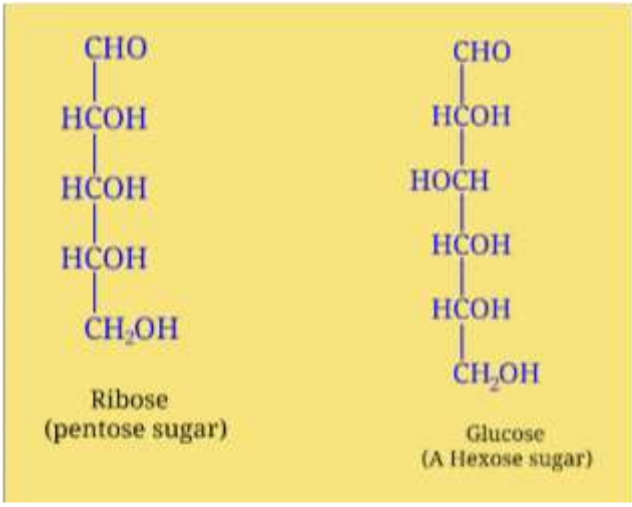
- Ribose is the component of RNA, ATP, NAD^+, FAD^+, $NADP^+$ etc.
- Ribulose is the component of RUBP which is the CO_2 acceptor in photosynthesis.

- Tetroses are rare in nature and ocuur in some bacteria.
- Most common are pentose and hexose. Most common hexose is glucose.



Glyceraldehyde

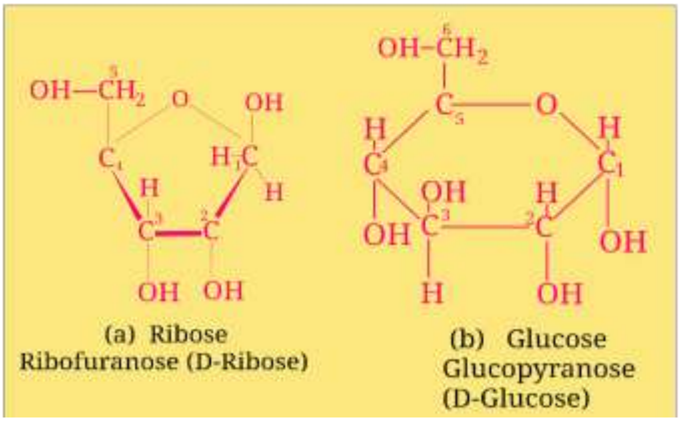
Dihydroxy acetone

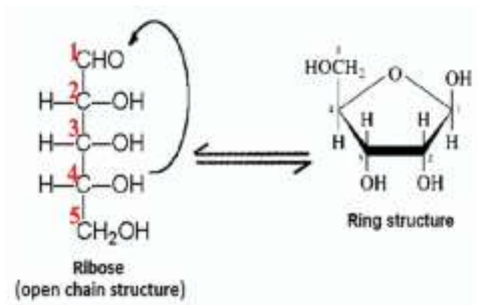


Important for first year MBBS as well:

Class	Aldoses	Ketoses	Functions
Trioses ($C_3H_6O_3$)	Glyceraldehyde	Dihydroxy acetone	Intermediate in photosynthesis and cellular respiration.
Tetroses ($C_4H_8O_4$)	Ethryose	Erythrulose	Intermediates in bacterial photosynthesis.
Pentoses ($C_5H_{10}O_5$)	Ribose, Deoxyribose ($C_5H_{10}O_4$)	Ribulose	Ribose and deoxyribose are components of DNA and RNA. Ribulose is an intermediate in Photosynthesis.
Hexoses ($C_6H_{12}O_6$)	Glucose, Galactose	Fructose	Glucose is respiratory fuel. Fructose is an intermediate in respiration. Galactose is component of milk sugar.
Heptoses ($C_7H_{14}O_7$)	Glucoheptose	Sedoheptulose	Intermediates in phoyosynthesis.

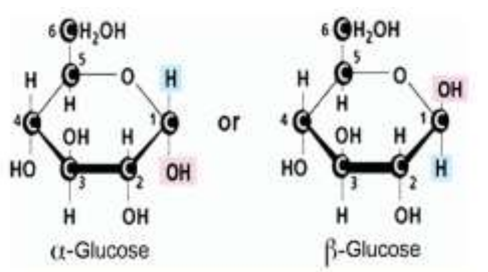
Chemical structures of monosaccharides





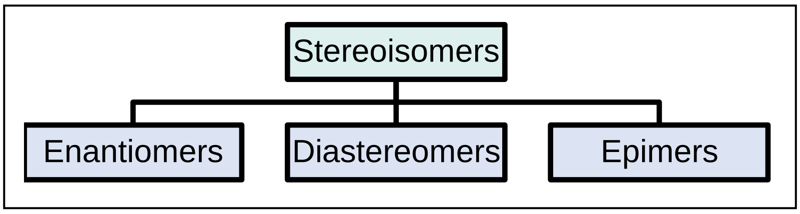
Interconversion of ring form and open chain structures. Fischer projection on the left and Haworth structure on the right.

- The open chain structure is known as **Fischer projection.**
- Normally these sugars are not found in open form in solutions.
- When they dissolved in water most of them (usually pentoses and hexoses) are converted into **ring chain structure.** This is known as **Haworth structure.**

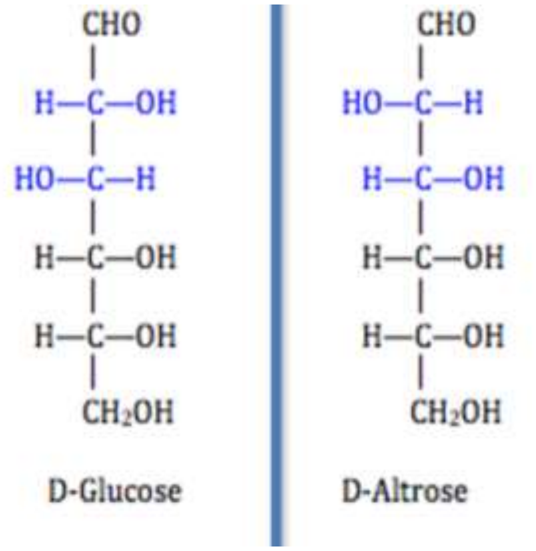


Anomeric Isomers of glucose

- For e.g. ribose will form a five cornered ring called **ribofuranose** and glucose will form a six cornered ring called **glucopyranose.**
- All pentoses and ketohexoses like fructose are converted into furanose ring.



- Second last carbon is called penultimate carbon e.g., carbon 4 in ribose.
- In ring structure formation,oxygen atom reacts with penultimate carbon to link.
- Let us consider example of ribofuranose.
- When it is dissolved in water, the oxygen atom from aldehyde group reacts with second last carbon i.e., C4 in case of ribose.
- In this way oxygen atom forms a link between C1 and C4 while the OH group of C4 is shifted to C1.
- Each pentose or hexose molecule in ring structure exists in either **α or β form** depending uponthe position of –H and -OH group on C-1.
- These are known as **Anomers**
  1. If -OH group is found **downward** on C-1 then it is called **α sugar**
  2. If-OH is present upward on C-1 then it is known as **β-sugar**.
- For example, consider ring structure of glucose.
- **Diastereoisomers** are sugars differeing in orientation at two carbons. (more than 1).
- For example, altrose is a diastereoisomers of glucose.
- **Epimers** are isomers of sugars that differ in the configuration at one carbon only.



Diastereoisomerism

- For example glucose and galactose differ at orientation at carbon no. 4 only

Enantiomers	Diastereomers	Epimers
These are non-superimposable image of each other.	These have different arrangement of H and –OH groups at more than one asymmetrical carbon atoms. These are not mirror images.	These have differed arrangement of H and OH groups at only one asymmetrical carbon are called epimers.
D and L isomers of glucose.	d-Glucose and D-altrose.	d-Glucose and d-mannose.

Comparison between structural isomers and sterisomers

- There are two forms of isomerism which are:

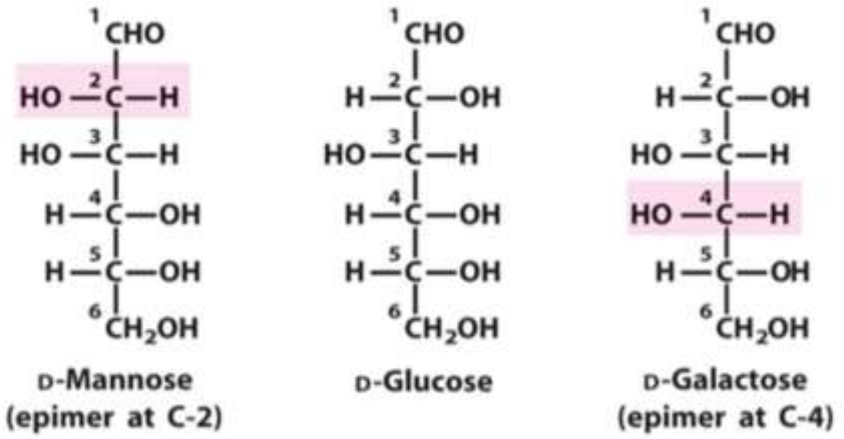
Structural isomers	Stereoisomers
In structural isomers (also called constitutional isomers) the atoms and functional groups are joined together in different ways.	In stereoisomers, the bond structure is same but the geometrical positioning of atoms and functional groups in space differs.
For example, glucose and fructose are structural isomers.	For example, D-glucose and L-glucose.

- Acetic acid,lactic acid and formaldehyde have the same formula as carbohydrates but they are not carbohydrates.
- While rhamnose $(C_6H_{12}O_5)_n$ does not match the carbohydrate formula but are carbohydrates.

Stereomerism in glucose

- Stereoisomers are molecules that have the same molecular formula and differ only in how atoms are arranged **in 3D space.**
- Those isomers in which -H and -OH groups are arranged in different pattern to the asymmetric carbon (chiral carbon) are called stereoisomers.
- An asymmetric carbon (chiral) is that which makes four bonds with different atoms.

- Stereoisomers are molecules that have the same molecular formula and differ only in how atoms are arranged in 3D space.
- In glucose, C2, C3, C4, C5 are asymmetric.
- In monosaccharides, the number of stereoisomers depend upon the number of asymmetric carbons and can be calculated by the formula 2n where n is number of asymmetric carbons.
- In glucose, there are 16 stereoisomers.



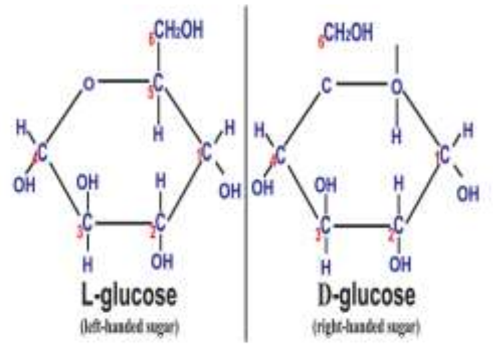
Epimers of glucose

(KPK)

- Stereoisomers are classified into 3 groups:
- Optical isomerism is one form of stereoisomerism.
- Optical isomers **(Enantiomers)** are two compounds which contain the same number and kinds of atoms and bonds (i.e., the connectivity between atoms is the same), and different spatial arrangements of the atoms, but which have non-superimposable mirror images.
- D-isomers (right handed form) are those in which asymmetric carbon of penultimate carbon has –OH group on right side.
- L-isomers (also called left handed form) the –OH group is on left hand side at penultimate carbon.
- Out of 16 stereoisomers of glucose, 8 are enantiomers of the other.
- Laboratory manufactured sugars are left handed.
- An example of enantiomer is D and L glucose. D sugars are right handed and L-sugars are left handed molecules.

Laboratory manufatured (artificial) sweeteners

- **Laboratory manufactured** sugars are **L** sugars. On the other hand the **naturally occurring** sugars in bodies are **D** sugars.
- Proteins and cell receptors are designed to reactonly with D sugars.
- For example, enzymes in your stomach can digest only right-handed sugars.
- A suitable analogy is pair of gloves, just like the glove fits only one on the proper hand, a right-handed enzyme cannot fit on or react with a left-handed substrate.
- The substrate must fit on the proper active site of the enzyme.
- So, for the left-handed substrate (artificial sweetner) the enzyme must be left-handed



An example of enantiomers.

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- The LH sugar have same physical properties as D-glucose, therefore, may be used instead of D-glucose e.g., for baking and also making ice cream.
- The left-handed sugars are not commonly used because they are expensive, not commonly available and their overuse cause serious disturbance for diarrhea patients.
- The laboratory manufactured sugar such as tagatose, sucralose etc. are examples of LH sugar.
- LH sugars are not converted into fats.

Occurrence of glucose

- In **free state**, glucose is present in **all fruits**, being abundant in grapes, figs, and dates.
- Grapes contain as much as 27% glucose.

- Our blood normally contains 0.08% glucose.

(KPK)

- Human blood contains 100 mg of glucose per 100 ml of blood. That's why glucose is known as **blood sugar.**
- In **combined form**, it is found in many disaccharides and polysaccharides.
- Honey contains large amounts of glucose and fructose.
- Starch, cellulose and glycogen yield glucose on complete hydrolysis only.
- Glucose is naturally produced in green plants through photosynthesis.
- For the synthesis of **10g** of glucose **717.6 kcal** of solar energy is used. This energy is stored in the glucose molecules as **chemical energy** and becomes available in all organisms when it is oxidized in the body.

Oligosaccharides

- These are comparatively less sweet in taste, and less soluble in water.
- On hydrolysis oligosaccharides yield from **two to ten** monosaccharides units
- The ones yielding two monosaccharides are known as **disaccharides**, those yielding three are known as **trisaccharides** and so on.

Characteristics	Sucrose	Maltose	Lactose
Common name	Cane sugar	Malt sugar	Milk sugar
Occurrence/ Sources	Sugar cane, sugar beet, home sugar, honey	Germinating seeds, barley, fruits, in our digestive tract as a result of starch digestion	4-6% in cow milk
Units	α - Glucose and β -fructose	Two alpha glucoses	Beta galactose and beta glucose
Linkage	α 1-2 glycosidic linkage between C1 of glucose and C2 of fructose.	α 1-4 glycosidic linkage between C1 of one glucose and C4 of other glucose.	β 1-4 glycosidic linkage between C1 of galactose and C4 of glucose.
Nature	Non-reducing sugar	Reducing sugar	Reducing sugar
Uses	used as **sweetener** at homes. In plants sucrose is also called **transport sugar** as prepared food in plants is transported in the form of sucrose because It is very soluble and also relatively **unreactive** chemically.	Used in **brewing** industry to synthesize alcohol. Maltose is an **intermediate** disaccharide produced during the breakdown of starch and glycogen.	It is an important **energy source** for young mammals. Used as **milk sweetener.**

- The covalent bond between two monosaccharides is called **glycosidic bond.**

Disaccharides

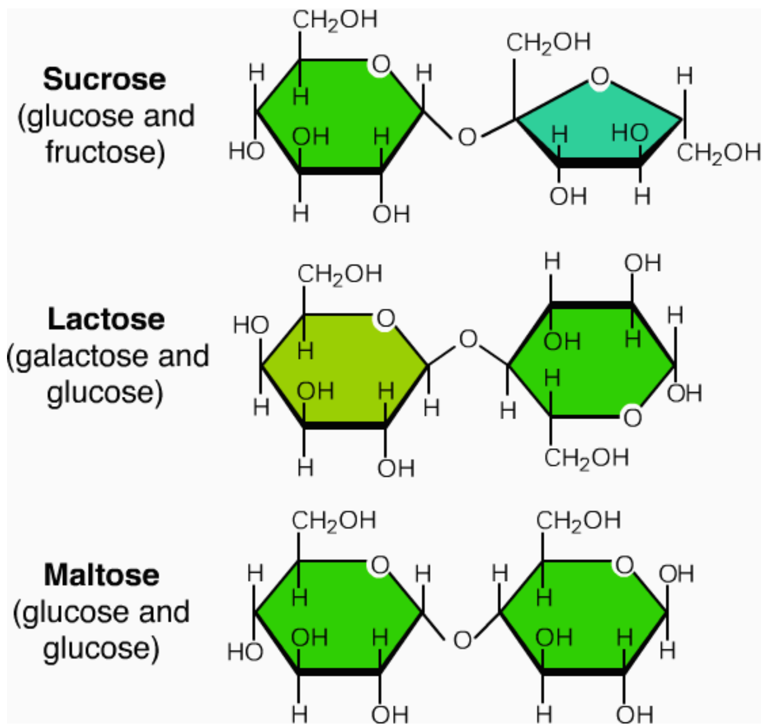
- **Most common** oligosaccharides are disaccharides.
- The general formula of disaccharides is $C_{12}H_{22}O_{11}$.
- Physiologically important disaccharides are maltose, sucrose, lactose.
- Most familiar disaccharide is sucrose (cane sugar) which on hydrolysis yields glucose and fructose.
- The sucrose is formed by the condensation of glucose and fructose. In this reaction, the -OH group at C-1 of glucose reacts with the -OH group at C-2 of fructose, liberating a water molecule forming a-1,2-glycosidic linkage.

(BTB)

- Maltose is found in our digestive tract as a result of breakdown product during digestion of starch by enzyme called amylase.
- It is also used in brewing industries to synthesize alcohol.
- In brewing industry, the maltose is produced from the breakdown of barley starch by the help of amylase enzyme.
- This is known as malting.

Reducing and Non-reducing sugars

- Any carbohydrate which is capable of being oxidized and causes the reduction of other substances without having to be hydrolyzed first is known as reducing sugar.
- All monosaccharides and two of three types of disaccharides (maltose and lactose) are **reducing sugars**.
- The third type of disaccharides, sucrose, and all polysaccharides are **non-reducing** sugars.



Structure of various disaccharides.

Polysaccharides

- Polysaccharides are the largest, the most complex and the **most abundant carbohydrates** in nature.
- They are usually branched and tasteless.
- They are formed by several (more than 10) monosaccharides linked by glycosidic bond.
- Polysaccharides have high molecular weights and are only insoluble or sometimes sparingly soluble in water.
- Polysaccharides function chiefly as food and energy stores and structural material.
- They are convenient storage molecule for several reasons.
- Their large size makes them more or less insoluble in water, so they exert no osmotic or chemical influence in the cell; they fold into compact shapes, and they are easily converted to sugars by hydrolysis when required.

Types of polysaccharides

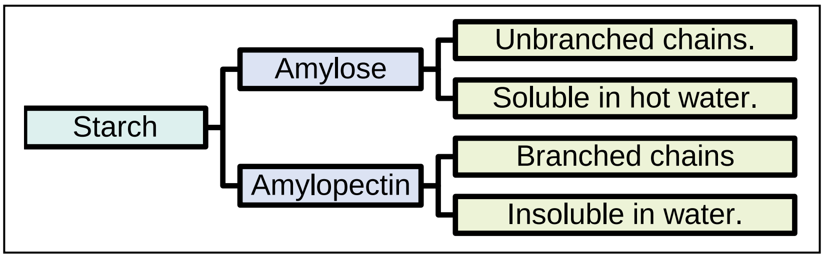
Polysaccharides can be classified as:

Homopolysaccharides	Heteropolysaccharides
The polysaccharides which are composed by the condensation of only one kind of monosaccharides are called homopolysaccharides.	The polysaccharide which are composed by the condensation of different kind of monosaccharides are called heteropolysaccharides.
e.g., starch, glycogen, cellulose, chitin.	e.g., agar, pectin, peptidoglycan.

Some common homopolysaccharides are discused here

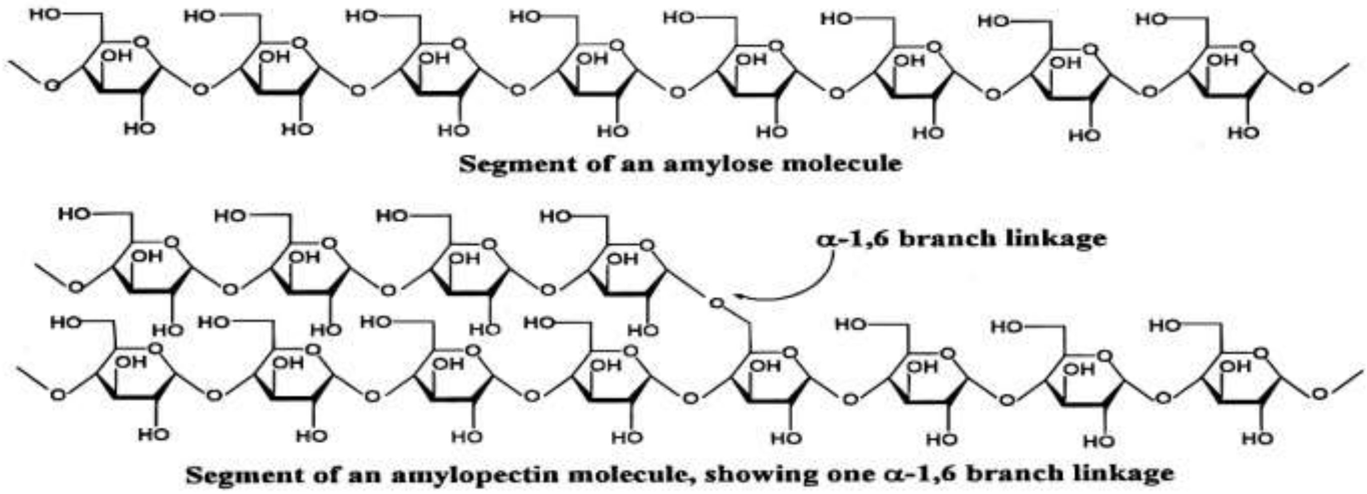
Starch

- Starch is formed by the condensation of hundreds of a-glucoses.
- It is called **storage carbohydrate** of plants. It is mainly stored in root, stem and seeds.
- It is found in roots, grains, seeds and tubers.
- Starch is **digested** in oral cavity and in small intestine by the enzyme amylase. Upon hydrolysis it yields maltose first and then maltose is further digested by maltase enzyme and yields glucoses.
- It gives **blue** colour with iodine solution.
- It is the main souce of carbohydrates in diet for animals.
- There are two **types** of starches i.e., amylose and amylopectin.



1. **Amylose:**

- It is **un-branched** i.e., a linear chain of glucoses in which glucoses are attached together by **α-1, 4-glycosidic linkages.**
- It is soluble in hot water and insoluble in cold water.



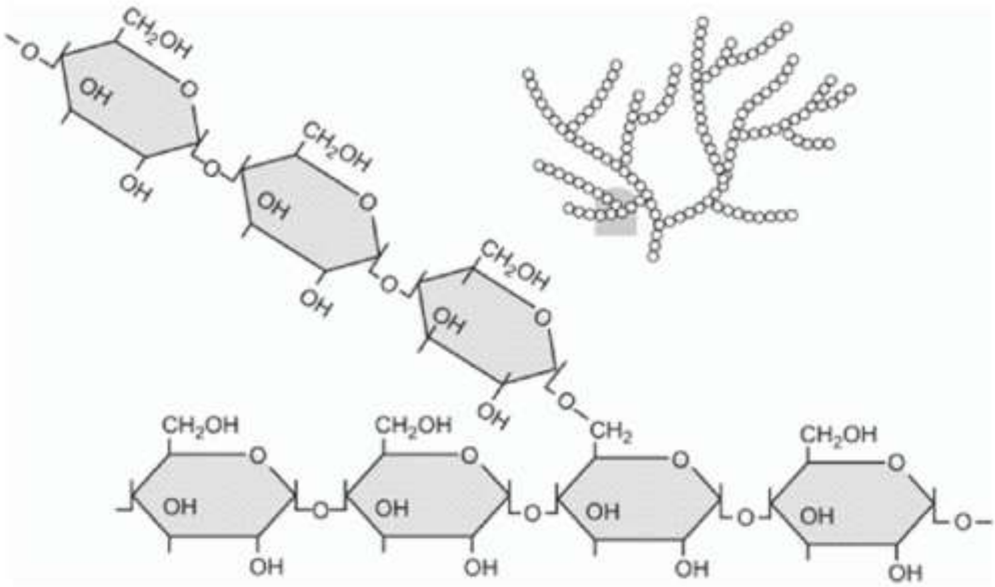
Structure of starches (Amylose and amylopectin)

2. Amylopectin:

- It has branched structure i.e., a linear chain of glucoses having **α-1,4 glycosidic linkages** but more chains of glucoses in the form of branches are also attached by **α-1, 6-glycosidic linkages**. (after 20 monomers, there is a branch).
- It is completely insoluble in both hot and cold water. **Mnemonic:** AmylopecT**I**n = **T**otally **I**nsoluble

Glycogen

- Glycogen is also composed of a-glucoses.
- It is called **storage carbohydrate of animals**. It is found abundantly in liver and muscles, though found in all animal cells. It is also found in fungi.
- It is also known as animal's starch.
- The **digestion** of glycogen is also quite similar to that of starch. It yeilds maltose upon initial hydrolysis and glucose after complete hydrolysis.
- It gives **red** colour with iodine solution.
- Structure of glycogen resembles with amylopectin starch but glycogen has much more **branching** than amylopectin (after 10 monomers,there is a branch)
- It is insoluble in water.
- Glycogen is also found in fungi.
- Glycogen has also alpha 1-4 and1-6 linkages.

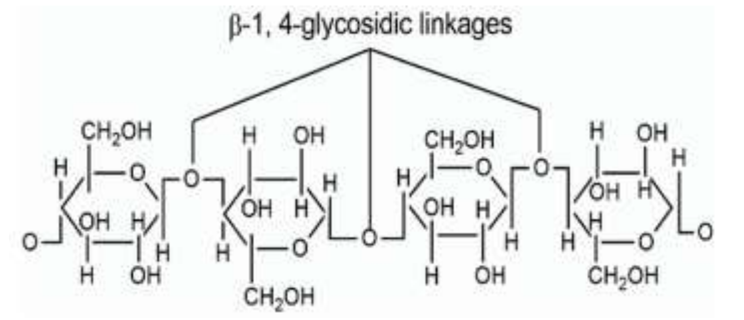


Structure of Glycogen

Cellulose (Fibre)

- Cellulose is the most abundant carbohydrate on earth.
- It is formed by condensation of hundreds of **β-glucoses**.

- It is **structural carbohydrate** of plants because it is the main constituent of their cell wall of plants (and algae also).
- Cotton and paper are pure forms of cellulose.
- It is not digested by human digestive tract. In herbivores, it is digested because of micro-organisms (bacteria, yeast and protozoa) in their digestive tract which secrete cellulase for digestion of cellulose.
- Upon intial hydrolysis, it yeilds cellubiose (a disaccharide with beta 1,4-glycosidic linkage) and after complete hydrolysis, it yeilds glucose.

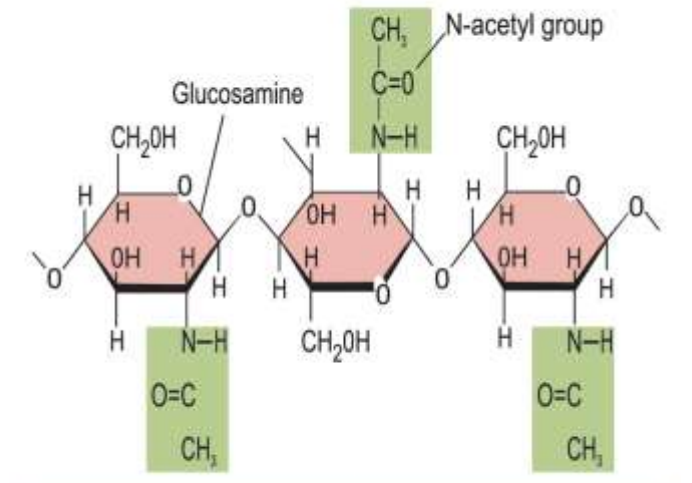


Structure of cellulose

- Cellulose cannot be digested by human body, but it has to be taken into diet because works as roughage or fibre, so it prevents abnormal absorption of food
- in intestine.
- However, herbivore animals have some symbiotic bacteria that secrete cellulase enzyme for its digestion.
- It gives **no** colour with iodine solution.
- Cellulose is added to iodine, the colour of iodine solution remains intact that is light orange brown [SZABMU MDCAT-I 2024]
- It is highly insoluble in water.
- Its structure resembles amylose starch as it is also unbranched but it has **β-1,4 glycosidic linkages** between glucoses.

Chitin

- Chitin is second most abundant organic molecule on earth.
- It is a derivative of **N-acetyl glucosamine** monomers.
- It is a structural nitrogenous polysaccharide found in cell walls of fungi and in the exoskeleton of arthropods.
- It is also known as **fungal cellulose**.
- Like cellulose, chitin is also undigestable.
- It is unbranched and it has **β-1,4 glycosidic linkage** between its monomers.



Structure of chitin.

Remember iodine test

Examples of homopolysaccharides (mnemonic = **CGS**)

- **C**ellulose — **CN** (No colour with iodine)
- **G**lycogen — **GR** (Red colour with iodine)
- **S**tarch — **BS** (Blue colour by starch)

Proteins

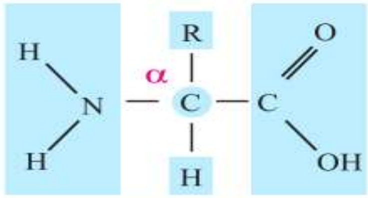
- Word "protein" has been derived from greek word **proteios** meaning prime or first.
- Proteins are the most abundant organic compounds to be found in cells and comprise over 50% of their total dry weight.
- One gram of protein gives **4.6 kcal** of energy.
- All proteins contain C, H, O and N, while some contains P, S. Few proteins have Fe, I and Mg incorporated into the molecule.
- Proteins are the main structural components of cell.
- Proteins may consist of single polypeptide or more than one polypeptide.
- Dipeptides and polypeptides are formed by condensation of amino acids on the ribosome under instructions of mRNA which takes these instructions from DNA. This process is known as translation.

Structure and Composition of proteins

- Chemically proteins can be defined as polymers of amino acids or polypeptide chains.
- The number of amino acids varies from **a few to 3000** or even more in different proteins.

Amino acids

- There are over 10,000 proteins in the human body which are composed of unique and specific sequence of 20 types of amino acids (characteristic feature of primary structure of a protein).
- About **170** types of amino acids have been found to occur in cells and tissues.
- Of these, about **25** are constituents of proteins.
- Most of the proteins are however, made of **20** types of amino acids.



General structure of an amino acid

(KPK)

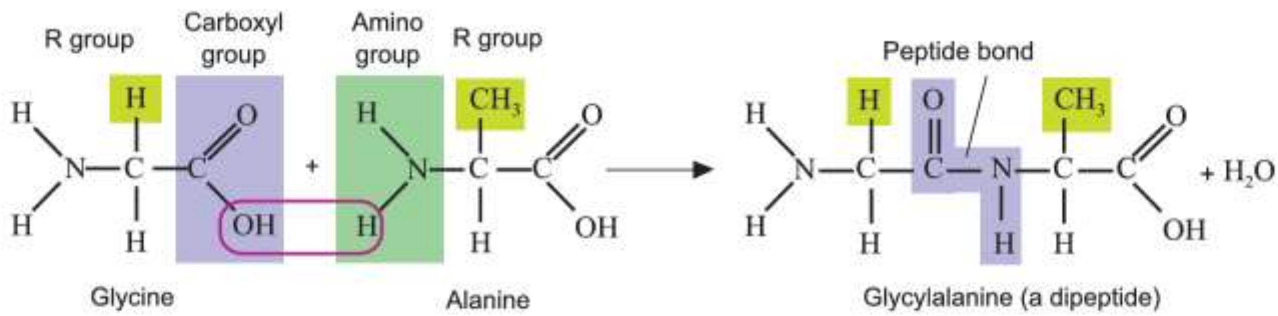
- Amino acids, the building blocks of proteins, are almost all left-handed--our bodies can't manufacture proteins out of the right-handed version.
- The cell walls of bacteria are one exception; they contain right-handed amino acids.

(BTB)

- Many amino acids are non-essential because body of the organisms can prepare them.
- Few amino acids are essential because body can't prepare them and are required in diet.
- All the amino acids have an amino group (-NH_2) and a carboxyl group (-COOH) attached to the same carbon atom also known as alpha carbon.
- The general formula of amino acids is:
- -R group is a variable group. It may be –H in glycine or –CH_3 in alanine.

Peptide linkage

- Amino acids are linked together to form polypeptides chains.The linkage between the hydroxyl group of **carboxyl group** of one amino acid and the hydrogen of **amino group** of another amino acid release H_2O and C-N link to form a bond called peptide bond.The resultant compound has two amino acid subunits and is a **dipeptide**.
- A dipeptide has two ends; one is called amino or **-N terminal** end while other is called carboxylic acid or **-C terminal** end.
- A new amino acid can be added in this chain from its carboxylic acid or -C terminal end in the same way.
- Similarly, when several amino acids are linked together by many peptide bonds, the polypeptide chain is formed.

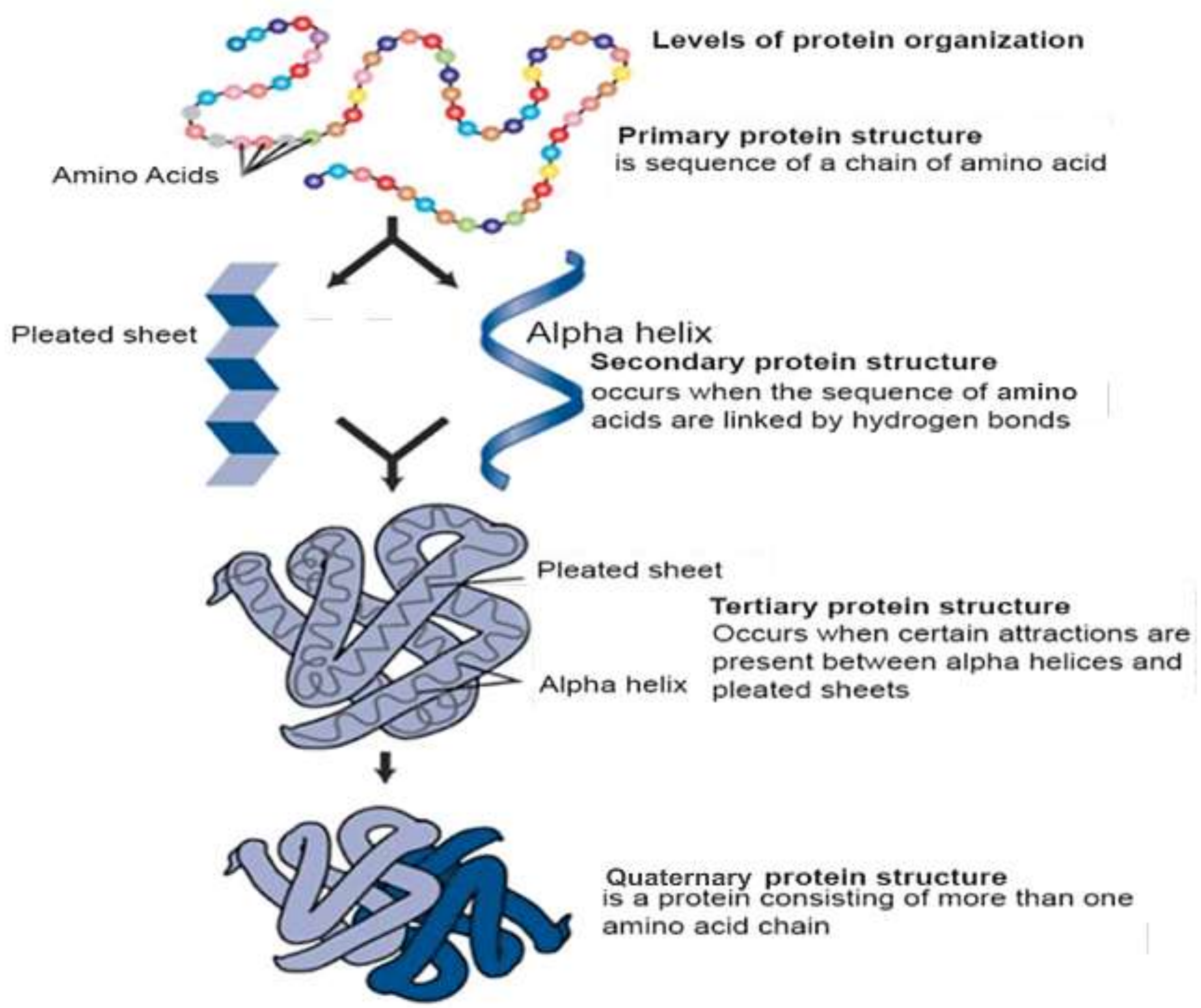


Formation of a dipeptide and peptide bond

Structure of proteins

There are four levels of organization of a protein molecule.

Protein organization
- Primary structure
- Secondary structure
- Tertiary structure
- Quaternary structure



Structural conformations in proteins

Primary structure

- The primary structure comprises the number and sequence of amino acids in a protein molecule.

(FTB)

- Primary structure is shown by all proteins at the time of their synthesis on ribosomal surface.
- F. Sanger in 1951 was the first scientist who determined the sequence of amino acids in a protein (insulin).
- **Insulin** is composed of 51 amino acids(49 peptide linkages) in two chains. One of the chains had 21 amino acids and the other had 30 amino acids and they were held together by disulphide bridges.
- Similarly, **Haemoglobin** is composed of 574 amino acids (570 peptide linkages) in four chains, two alpha and two beta chains. Each alpha chain contains 141 (282 in both) amino acids, while each beta chain contains 146 amino acids (292 in both).

- Insulin
  - 21 amino acids (α-chain)
  - 30 amino acids (β-chain)

- The size of a protein molecule is determined by the type and number of amino acids comprising that particular protein molecule.

- Hb (574 amino acids)
  - Two alpha chains
    - 141 amino acids
    - 141 amino acids
  - Two beta chains
    - 146 amino acids
    - 146 amino acids

- To calculate the number of possible peptides of a given length from a set of amino acids, you use the formula: **Number of peptides = (number of amino acids)$^{\text{(length of peptide)}}$**
- For example, Number of different dipeptides from five different amino acids is $5^2 = 25$ dipeptides.

Secondary structure

- The polypeptide chains then usually coil into α-helix, or into some other regular configuration which is a secondary structure.
- One of the secondary structures, **α-helix** involves a spiral formation of polypeptide chain.
- It is a very uniform geometric structure with **3.6** amino acids in each turn (36 in 10 turns) of the helix.

In chemistry, select option that *there are 27 amino acid units for each turn of the helix.*

- The helical structure is kept by the formation of **hydrogen bonds** among amino acid molecules in successive turns of the spiral.
- **β-pleated sheet** is formed by folding back of the polypeptide.

Tertiary structure

- Usually a polypeptide chain **bends and folds upon itself** forming a globular shape or **3-D structure**. This is the proteins' tertiary conformation.
- It is maintained by three types of bonds, namely ionic, hydrogen, and disulfide (-S-S-).
- Disulphide bridges are present in antibodies and are formed by the presence of cysteine.
- **Metal ion coordination** can be seen from tertiary structure onwards, e.g Fe^{2+} as heme group.
- In aqueous environment the most stable tertiary conformation is that in which hydrophobic amino acids are buried inside while the hydrophilic amino acids are on the surface of the molecule.

Explanation:

- In an aqueous environment (water-based environment, such as inside the body), proteins fold into a tertiary structure that is most stable and energetically favorabl
- This folding follows the hydrophobic effect, which drives:
- Hydrophobic (nonpolar) amino acids to be buried inside the protein structure, away from water
- Examples: Valine, Leucine, Isoleucine, Phenylalanine
- This reduces their interaction with water, increasing stability.
- Hydrophilic (polari amino acids to be exposed on the outside, interacting with water
- Examples Serine, Threonine, Glutamine, Lysine
- These form hydrogen bonds with water, stabilizing the structure.
- This arrangement minimizes the free energy of the protein and ensures proper function and say in the aqueous environment

Quarternary structure

- When **two or more polypeptide** tertiary chains are aggregated and held together by hydrophobic interactions, hydrogen and ionic bonds, the specific arrangement is the quaternary structure.
- For e.g, hemoglobin.

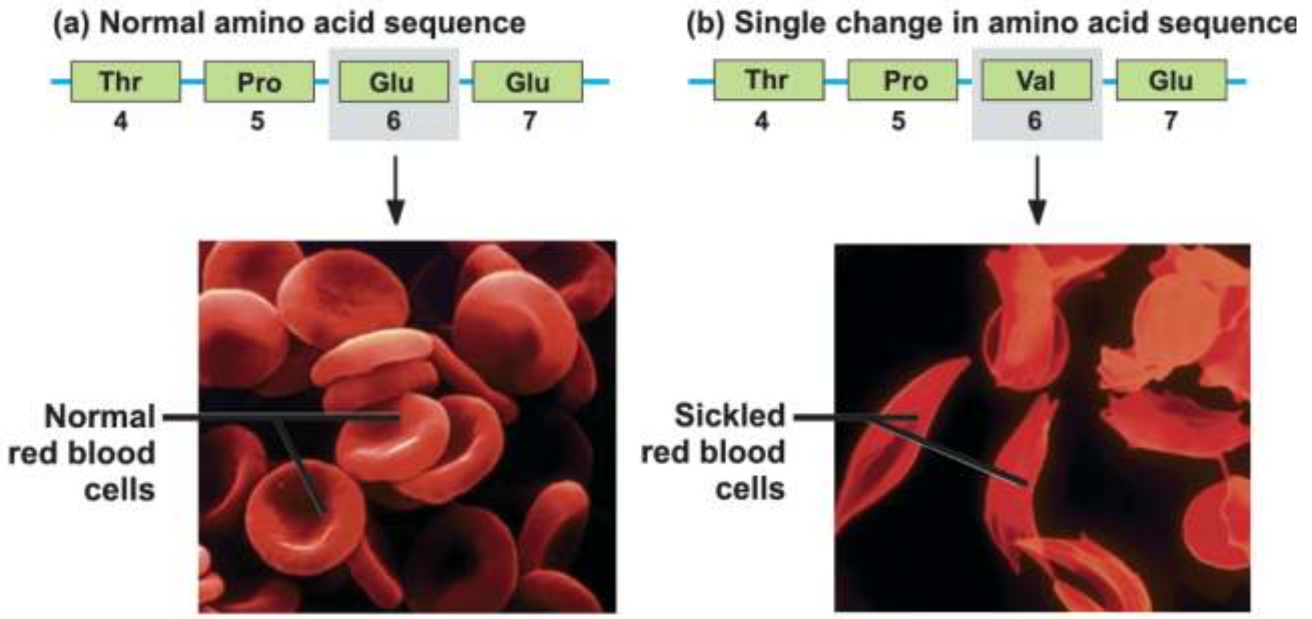
Quaternary structure

Hydrogen bond. | Ionic bonds | Hydrophobic interactions.

Significance of sequence of amino acids

- The sequence of amino acids in a protein is determined by order of nucleotides in DNA.
- The arrangement of amino acids in a protein molecule is highly specific for its proper functioning.
- The best example is the **sickle cell hemoglobin** of human beings.
- Normal RBCs are disc-shaped and look like doughnuts without holes in the centre.
- But in case of sickle cell anemia, body makes sickle or crescent shaped RBCs. They contain abnormal hemoglobin called sickle haemoglobin (Hb^S).
- Sickle cell anemia is caused by a mutation **(point mutation)** and type of mutation is substitution. In β-globin gene in which only one nucleotide is replaced by another which causes a change in amino acid sequence of both β-chains of haemoglobin.
- Sickle cell Hb^S shows only one difference from normal Hb^A i.e., polar glutamic acid is replaced by non-polar valine at position number six in both β-chains.

- Hence, hemoglobin fails to carry any or sufficient oxygen, hence leading to death of the patient.



Point mutation in sickle cell haemoglobin

Classification of proteins

Fibrous proteins	Globular proteins
They consist of molecules having one or more polypeptide chains in the form of fibrils or filaments.	**1.** These are spherical or ellipsoidal or globular due to multiple folding of polypeptide chains.
Secondary structure is most important in them.	**2.** Tertiary structure is most important in them.
They are insoluble in aqueous media.	**3.** They are soluble in aqueous media such as salt solution, solution of acids or bases, or aqueous alcohol.
They are non-crystalline and are elastic in nature.	**4.** They are crystalline and inelastic. **5.** (**mnemonic:GC**= **G**lobular **C**rystalline)
They perform structural roles in cells and organisms.	**6.** They disorganize with changes in the physical and physiological environment.
Examples Silk fiber (from silk worm, spiders web) Actin and myosin (in muscle cells) Fibrin (of blood clot) Keratin (of nails, hair, fur, outer skin) Collagen (in skin, ligaments, tendons, bones and in cornea of eyes).	**Examples** Enzymes, antibodies, hormones, hemoglobin, myoglobin, channel proteins, albumen of egg white and proteins of cell membranes.

Note: Myosin, a fibrous protein has enzymatic activity due to its globular heads.

Role of proteins

Some proteins have structural roles while some have functional roles.

- Following is the list of **structural proteins**.

Types	Roles of proteins
Collagen	It establishes the matrix of bones and cartilage.
Elastin	Elastin provides support for connective tissues such as bones and cartilage.
Keratin	It strengthens protective coverings such as hair, nails, quils, feathers, horns and beaks.
Histone	It arranges the DNA into the chromosome.

- Following is the list of **functional proteins**.

Types	Roles of proteins
1. Enzymes	Most enzymes are proteins which control metabolism i.e., they speed up biochemical reactions.
2. Hormones	Some hormones are protein in nature which are involved in regulation of physiological activities.
3. Antibodies	These proteins are produced by WBCs in response to antigens and provide immunity.
4. Haemoglobin	It is found in RBCs and is involved in the transport of oxygen mainly and carbon dioxide to some extent.
5. Fibrinogen	It is found in blood plasma and is involved in the blood clotting process.
6. Ovalbumin and casein	Ovalbumin is found in egg whites and casein is a milk based protein.

Most abundant proteins

- The most **abundant** protein:

1. Earth = rubisco **2.** Human = collagen type (I) **3.** Blood = albumin

(KPK)

- In plants, proteins are stored in most seeds for future need of the embryos. e.g., bean, pulses, pea etc.

(BTB)

Other functions

- Myoglobin is another protein complex that stores oxygen in the red muscles.
- Protein molecules also store energy in muscles of the body which supply energy to the body when outside source of food is Inadequate like phosphocreatin
- Prothrombin is also a blood clotting protein.
- - Contractility is one of the most outstanding property of proteins.
- Contractile muscle proteins (actin and myosin).

(PTB)

- Tubulin of microtubule (cilia, flagella and centrioles) help in the movement of chromosomes during anaphase caused by proteins (spindle fibers)
- Movement of organs and organisms, and movement of chromosomes during anaphase of cell division, are caused by proteins.

Lipids

- Lipid is the collective name for variety of organic compounds such as fats, oils, waxes and fat-like molecules (steroids) found in the body.
- Therefore, it is defined as a **heterogeneous group** of organic compounds.
- Most lipids are non-polar and are insoluble in water (hydrophobic) but soluble in organic solvent such as acetone, alcohol, and ether, benzene chloroform etc.
- Lipids are composed of carbon, hydrogen and oxygen. Nitrogen and hosphorous may also be present.
- However, they have relatively less oxygen in proportion to carbon and hydrogen.
- The percentage of oxygen in lipids is less than the carbohydrates which makes lipids lighter and make it much less soluble in water than most carbohydrates.
- For instance, tristearin is a simple lipid which shows molecular formula as $C_{57}H_{110}O_6$.
- Due to high contents of carbon and hydrogen, they store double amount of energy than carbohydrates.
- One gram of lipid gives **9 kcal** of energy.
- They play an important role in structure of membranes of cells and its organelles.
- **Bloor** proposed the term Lipid in 1943.

Functions of lipids

- Lipids are components of cellular **membranes** (e.g, phospholipids and cholestrol).
- They act as **energy stores** (e.g, triglycerides).
- Some **hormones** are lipids For e.g., auxins, gibberellins and cytokinins among plant hormones and aldosterone and sex hormones among animal hormones.

- Lipids are also involved in mechanical protection, insulation, waterproofing and buoyancy.
- Waxes, in the exoskeleton of insects. and cutin, an additional protective layer on the cuticle of epidermis of some plant organs e.g. leaves, fruits, seeds etc., are some of the main examples.

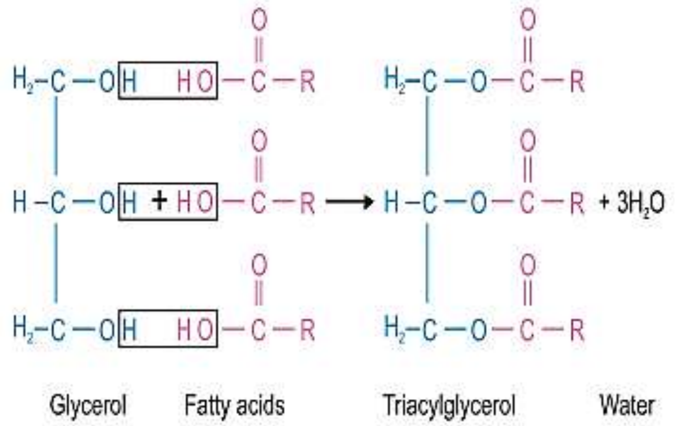
Functions of lipids
- Protection
- Water proofing
- Insulation
- Buoyancy

Classification of lipids

- Lipids have been classified based on their solubility and products of hydrolysis as acylglycerols, waxes, phospholipids, sphingolipids, glycolipids and terpenoid lipids.

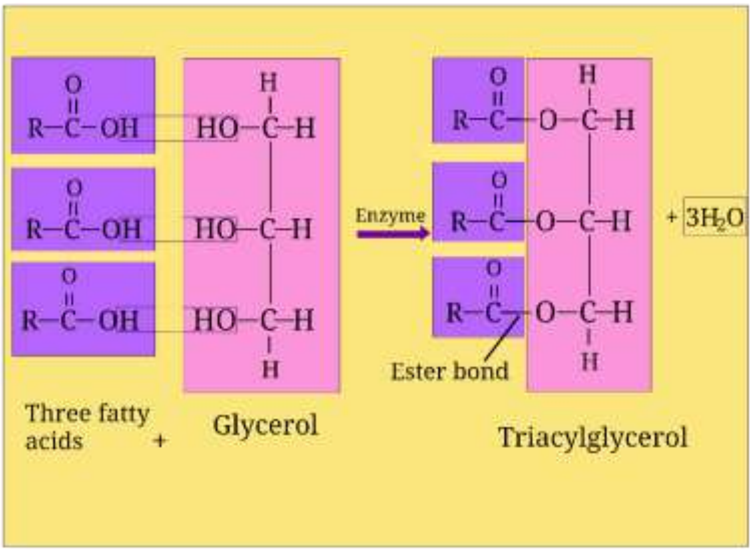
Classification of lipids
- Acyl-glycerols
- Waxes
- Phospho-lipids
- Terpenes
- Sphingo-lipids
- Glyco-lipids
- Steroids
- Prosta-glandins

Acylglycerols



Formation of a triacylglycerol (in the presence of H^+ catalyst) - FTB

- The most abundant lipids in living things are acylglycerols.
- Acylglycerols can be defined as **esters** of glycerol (alcohol) and fatty acids.
- An ester is the compound produced as the result of a chemical reaction of an alcohol with acid and a water molecule is released such a reaction is called esterification.
- **Glycerol** is a trihydroxy alcohol which contains three carbons, each bears an OH group.
- A **fatty acid** is a type of organic acid containing one carboxylic acid group attached to a hydrocarbon.
- Fatty acids contain even number of carbons from 2 to 30.
- Each fatty acid is represented as R-COOH, where R is a hydrocarbon tail.
  1. When one glycerol molecule combines chemically with one fatty acid, a **monoacylglycerol** (monoglyceride) is formed.
  2. When two fatty acids combine with one glycerol a **diacylglycerol** is formed.
  3. Similarly, three fatty acids + one glycerol molecule = **triacylglycerol**.
- These are most common acylglycerols.
- Triacylglycerols are also called **neutral lipids** as all three charge bearing OH groups are occupied by three fatty acids.
- Acylglycerols are called neutral lipids because both acid and base are present in them.



In the presence of enzyme catalyst - PTB

$$\underset{\text{alcohol}}{C_2H_5OH} + \underset{\text{acetic acid}}{HOOCCH_3} \longrightarrow \underset{\text{an ester}}{C_2H_5OCOCH_3} + H_2O$$

(BTB)

- After esterification, there are no free carboxyl (-COOH) or hydroxyl (-OH) groups left to ionize, making the molecule uncharged (neutral)

(KPK)

- During formation of triglycerides, three water molecules are released and process is called condensation.
- The most widely spread acylglycerol is triacylglycerol.
- In the formation of acylglycerols, OH is released from alcohol and H from acid which are together released as water molecule (condensation) . (But according to FTB, OH is taken from acid and H from alcohol).

Reason: In laboratory/in vitro/chemistry, OH comes from acid because there is H^+ catalyst and it is nucleophilic substitution of carboxylic acid. But in vivo/in cell/in biology, ther reaction is carried out by enzymes and in its mechanism, enzyme removes OH from alcohol and H from acid.

Acylglycerols
- Monoacylglycerol → Glycerol + 1 fatty acid
- Diacyglycerol → Glycerol + 2 fatty acids
- Triacyglycerol → Glycerol + 3 fatty acids

Fatty acids properties

(KPK)

- Fatty acids contain even number (2-30) of carbon atoms in a straight chain attached with hydrogen and have an acidic group (carboxylic group)
- Each fatty acid is represented by RCOOH, where R is hydrocarbon tail.
- About **30** different fatty acids are found.
- Most important component of Triglycerides.
- Acetic acid (2C) and butyric acid (4C) are simplest fatty acid.
- Palmitic acid (16C) and stearic acid (18C) are most common fatty acids.
- Some properties of fatty acid are increased with an **increase in number of carbon atoms**, such as melting point, solubility in organic solvent and hydrophobic nature.
- For e.g., Palmitic acid (C_{16}) is much more soluble in organic solvents and has higher melting point than butyric acid (C_4).
- Oleic acid is unsaturated fatty acid (double bond is present between C_9 and C_{10}).
- Most fatty acids in **plants** contain 16-18 carbon per molecule
- In animals the fatty acids are straight chains, while in plants these may be branched or ringed.
- Fatty acids may be saturated or unsaturated.

Saturated fatty acids	Unsaturated fatty acids
1. They contain no double bond.	**1.** They contain upto six(1-6) double bonds.
2. Straight chain	**2.** Ring or branched
3. Saturated fatty acids tend to be solid at room temperature.	**3.** Unsaturated fatty acids tend to be liquid at room temperature.
4. They have higher melting point.	**4.** They have lower melting point.
5. They contain more energy due to more C-H bonds.	**5.** They contain less energy due to less C-H bonds.
6. They are more common in animal lipids which are called **fats.**	**6.** They are more common in plant lipids which are called **oils.**
Exapmles: Lauric acid, myristic acid, palmitic acid, stearic acid.	**Examples:** Oleic acid, linoleic acid, linoleneic acid, arachidonic acd.

- Fats and oils are lighter than water.
- These have specific gravity of 0.8.
- They are non-crystalline but some can be crystallized under specific conditions.

(BTB)

- Ghee with saturated fatty acids is prepared from vegetable oil by passing hydrogen through it.
- Intake of ghee should be minimized as it may store in blood vessels reducing their flow capacity increasing risk of heart attack
- Some common fatty acids are given in following table.

Name	Typical Source	No. of Carbon	Condensed Formula	Melting Point (°C)	Type
Acetic acid	Vinegar	2	CH_3COOH	16.6	Saturated
Palmitic	Most fats and oils, Palm tree	16	$CH_3(CH_2)_{14}COOH$	63.1	Saturated
Stearic	Most fats and oils	18	$CH_3(CH_2)_{16}COOH$	70	Saturated
Butyric acid	Butter and dairy products.	4	$CH_3(CH_2)_2COOH$	-8	Saturated
Oleic	Olive oil	18	$CH_3(CH_2)_7CH=CH(CH_2)_7COOH$	4	Monounsaturated
Linoleic	Vegetable oils	18	$CH_3(CH_2)_4CH=CHCH_2CH=CH(CH_2)_7COOH$	-5	Polynsaturated

Note

- Stearin is found in beef and mutton.
- Linolin ($C_{57}H_{104}O_6$) is found in cotton seed have linoleic acid.

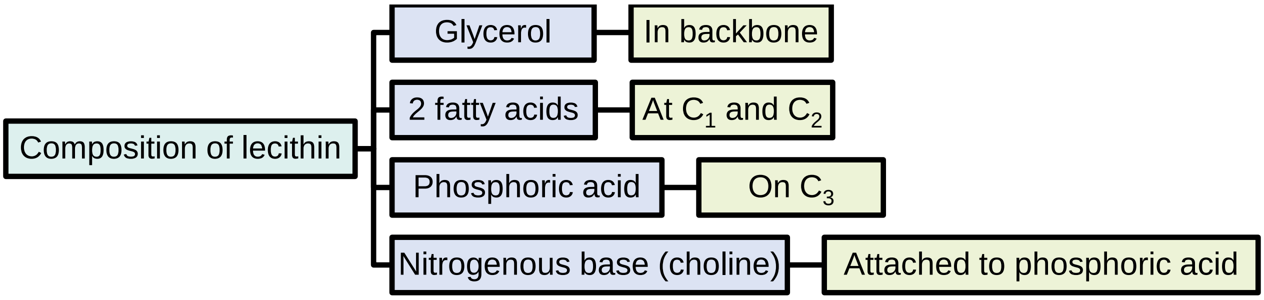
Waxes

- Waxes are mixtures of long chain alkanes (with odd number of carbon atoms ranging from C_{25}-C_{35}) and alcohols, aldehydes, ketones and esters of long chain fatty acids.
- Waxes are highly hydrophobic compounds.
- There are two types of wax.
- **Natural waxes** are simple lipids.
- They are typically esters of long chain fatty acids and long chain alcohols.
- They are solid at room temperature because they have high melting point.
- They are stable compound/inert and resistant to degradation and atmospheric oxidation.
- They are widespread as protective coatings on fruit and leaves and protect plants from water loss and abrasive damage.
- Some insects also secrete wax.
- They also provide water barrier for insects, birds and animals like sheep.
- For example:
  1. bees wax (the most common animal wax and found in honeycomb)
  2. cutin (on leaf surfaces and fruits) of plants)
  3. suberin (in cell wall of endodermis of plant roots)
  4. lanolin (found in sheep wool)
- **Synthetic waxes** which are generally derived from petroleum or polyethene.
- For e.g., paraffin wax which is used to make candles, wax paper, lubricants and sealing material.
- Formula of wax is $CH_3(CH_2)_4COO(CH_2)_{29}CH_3$.
- The most common plant wax is **epicuticular wax**.

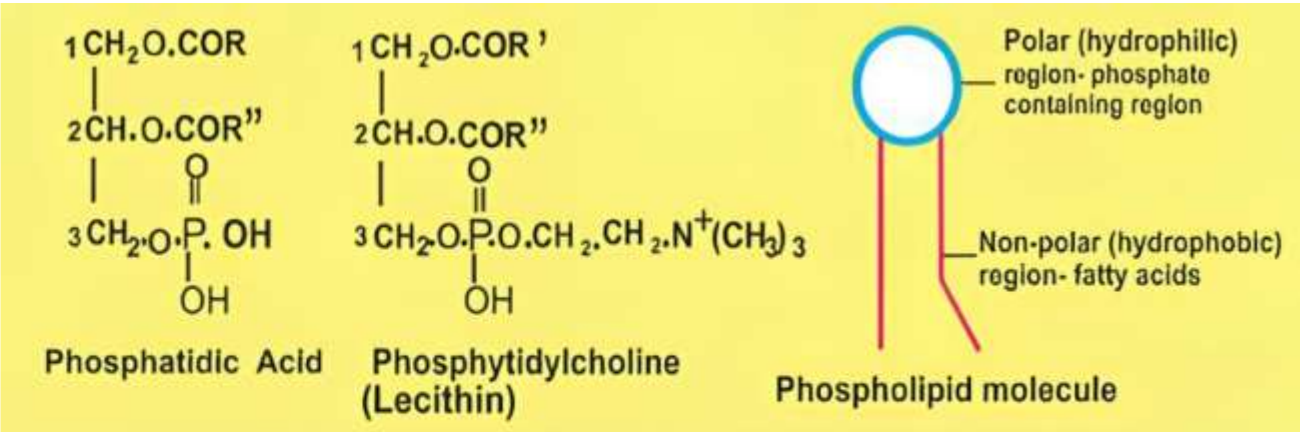
Phospholipids

- These are the most abundant lipids.
- Phospholipids are derivatives of **phosphatidic acid**.
- A phosphatidic acid molecule is most similar to diglyceride that it contains **a glycerol**, **two fatty acids** esterified with first and second OH groups of glycerol and **a phosphate** group esterified with third OH group of glycerol.
- Nitrogenous bases are important components of phospholipids.

- Phospholipid molecule contains two parts.
  - ❖ 3 fatty acids + 1 glycerol = triglycerides.
  - ❖ 2 fatty acids + 1glycerol + phosphate group = phosphatidic acid
  - ❖ 2 fatty acids + 1glycerol = phosphate group + N-base = phospholipids.
- A phospholipid is formed when phosphatidic acid combines with one of the four organic compounds such as :
  1. choline (a nitrogenous base) **2. ethanolamine** (an amino alcohol)
  3. inositol (an amino alcohol) **4. serine** (an amino acid).
- Most common type of phospholipid is **phosphatidylcholine** also called lecithin in which choline is attached to phosphate group of phosphatidic acid.

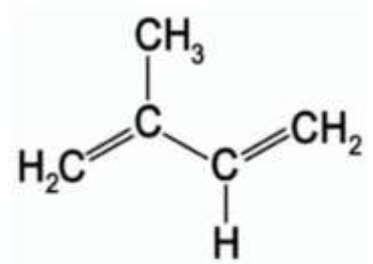


- Phospholipids are major compononets of cell membranes in the form lipid bilayer because they are **amphiphilic**.
- One end of the phospholipid molecule, containing the phosphate group and additional compound (head region) is hydrophilic i.e, polar and readily soluble in water.
- The other end, containing the fatty acid side chain (tail region) is hydrophobic i.e. non-polar.
- They arrange
- All lipids are hydrophobic except phospholipids head.
- Phospholipid protein complex are present in milk, blood, cell nucleus and eggs.



Terpenoids

- Terpenoids are a very large and diverse group of organic compounds.
- They are the only polymers among the class of lipids and are non-fatty acid lipids.
- The building block of terpenoids is a five-carbon **isoprenoid unit.**
- **Classification of terpenoids:** Isoprenoid unit by condensation in different ways give rise to compounds such as **rubber, carotenoids, steroids and terpenes**.



Isoprenoid unit

Terpenes

- These are polymers of isoprene units. C_5H_8
- Two isoprene units form a **monoterpene** ($C_{10}H_{16}$) e.g, menthol, four form a **diterpene** ($C_{20}H_{32}$) e.g., vitamin A, phytol (chlorophyll tail) and six form a **triterpene** ($C_{30}H_{48}$) e.g., ambrein.
- Produced by a variety of plants and some insects as well.
- Natural rubber is a **polyterpene**.
- Terpenes help in oxidation-reduction reaction.

- Terpenes are derived lipids.

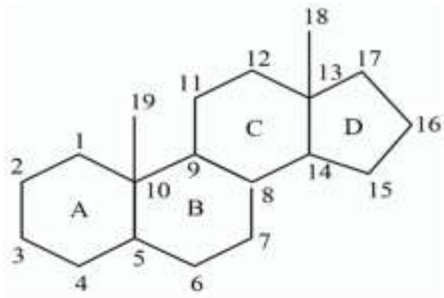
(FTB)

- Isoprene unit is a 5 carbon unsaturated compound (C_5H_8) (2-Methyl-1,3-butadiene)

Monoterpene (2 isoprene units)	Menthol
Diterpene (4 isoprene units)	Vitamin A, phytol (chlorophyll tail)
Triterpene (6 isoprene units)	Ambrein
Polyterpene	Natural rubber

Steroids

- Steroids are non-fatty acids lipids of high molecular weight which can be crystalline.
- A steroid nucleus consists of **17** carbon atoms arranged in **four** attached rings.



Steroid Nucleus

(FTB)

- Three of the rings are six sided, and the fourth is five sided.
- These structures are synthesized from isoprene units.
- The length and structue of side chains distinguish different types of steroids.
- Example, **cholesterol** is the structural component of cell membrane and brain tissue.
- **Sex hormones** like estrogen, progesterone in female and testosterone in male are steroids in nature derived from cholesterol.
- **Vitamin D** which regulates calcium metabolism and **bile salts** which emulsify fats are steroids.

Prostaglandins

- The name prostaglandins is derived from prostate gland because it was first isolated from seminal fluid in 1935.
- It was believed to be part of prostatic secretions.
- They are found in every mammalian tissue and act as local hormones.
- They sensitize spinal neuron for pain.
- Thermoregulatory centre of hypothalamus to regulate fever.
- They are derivatives of **arachidonic acid.**
- Every prostaglandin contains **20** carbon atoms including a 5-carbon ring.
- Aspirin like drugs inhibit synthesis of prosatglandin.
- Their function vary widely depending on the tissues
- Some reduce blood pressure others raise it.
- In the immune system various prostaglandins help to induce fever and inflammation and also intensify the sensation of pain.
- They also help to regulate the aggregation of platelets, an early step in the formation of blood clots.
- Infact the ability of aspirin to reduce fever and decrease pain depends on the inhibition of prostaglandins synthesis.

Uses of synthetic prostaglandin

- Induce parturition
- Prevent peptic ulcer
- Treat peptic ulcer
- Prevent egg binding
- Treatment of pulmonary hypertension

Nucleic acids

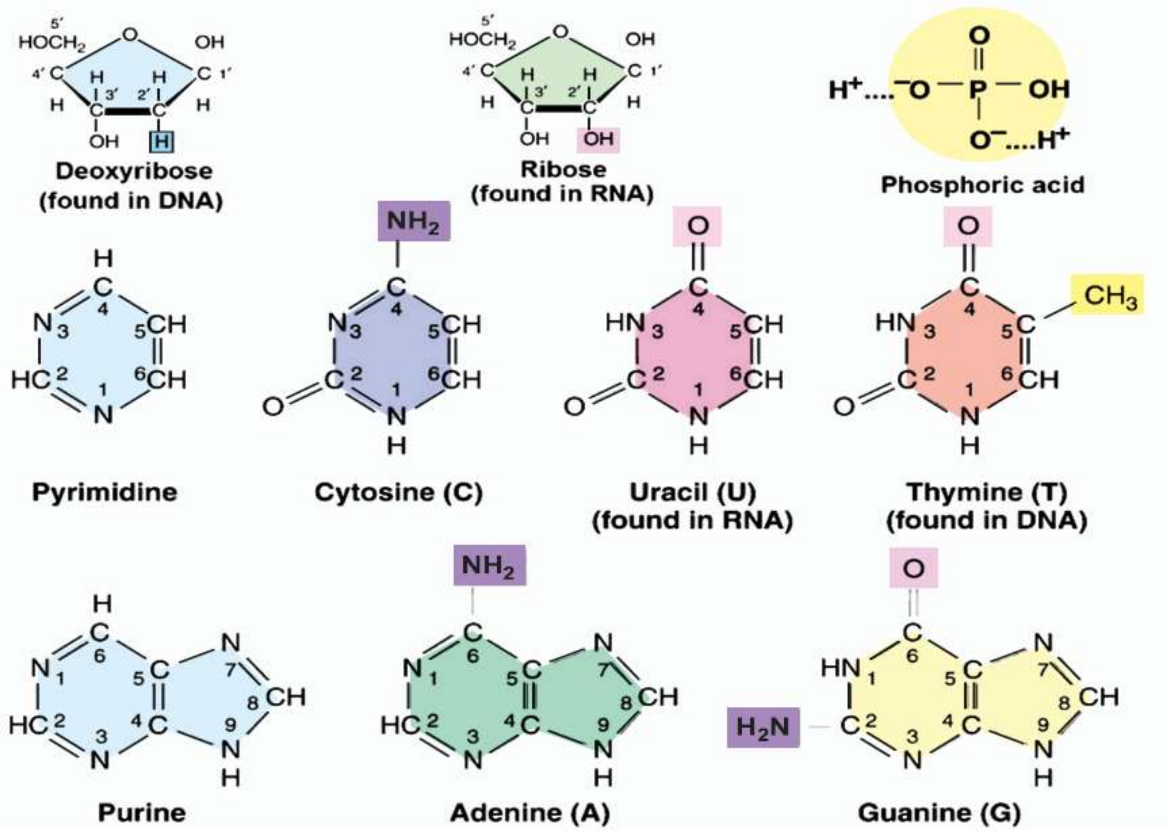
- Nucleic acids were first isolated (in 1869) by a Swiss physician, Fredrick miescher from the nuclei of **pus cells** (white blood cells) and **sperm** of salmon fish.
- He named this molecule as nuclein, because it was located in the nucleus.
- The basic structure and chemical nature of nuclein was determined (in 1920) and was renamed as nucleic acid because of its acidic nature.
- Nucleic acids are of two types, **DNA and RNA.**

- Nucleic Acids
  - DNA contain deoxy ribose
  - RNA contains ribose
    - tRNA transfers Amino acids
    - rRNA catalytic part of ribosomes
    - mRNA transfers information about amino acid sequence

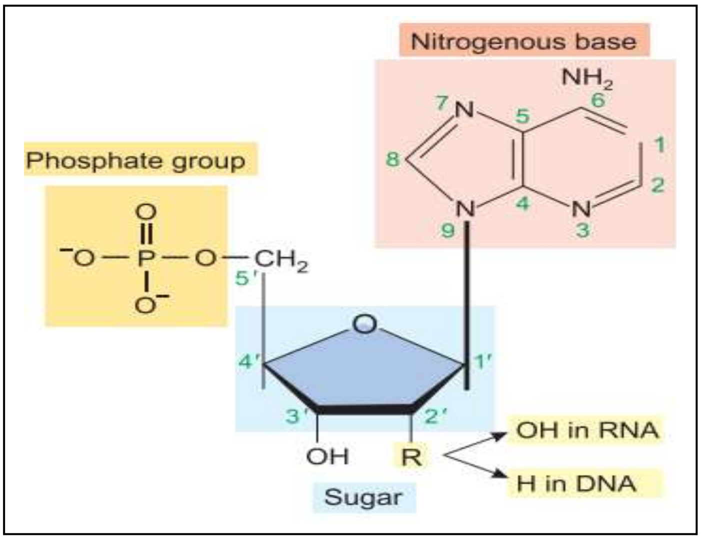
- DNA occurs in chromosomes, in the nuclei of the cells and in much lesser amounts in mitochondria and chloroplasts. RNA is present in the nucleolus, in the ribosomes, in the cytosol and in smaller amounts in other parts of the cell.
- Both are linerar, unbranched polymers of **nucleotides.**
- DNA is stable long-term protein and nucleic acid but RNA degrades within 30 minutes in cell.

Constituents of nucleotide

- The partial hydrolysis of nucleic acids yield compounds known as nucleotides or nucleosides.
- While, complete hydrolysis yields a mixture of bases, pentose sugars and phosphate ions.
- Nucleotides of DNA are **deoxyribonucleotides** while that of RNA are **ribonucleotides.**
- Both nucleotides consists of **three parts**
  1. a pentose sugar **2.** a phosphate
  3. a nitrogen containing ring structure called base.
- The **pentose sugar** in deoxyribonucleotides is deoxyribose and in ribonucleotides is ribose.
- Deoxyribose has same structure as ribose except it has one oxygen removed from OH at carbon 2.



- Di-deoxyribonucleoside triphosphates are used to terminate DNA synthesis at different sites in Sanger's method.
- **Phosphoric acid** is a common component of both nucleotides which provides acidic properties to DNA and RNA.
- The nitrogen containing heterocyclic structures are called **bases** because of unshared pair of electron on nitrogen atoms, which can thus acquire a proton.
- There are two major **classes of nitrogenous bases** i.e., single six cornered ring, pyrimidine and double ring (a 6 cornered ring attached to a 5 cornered ring)purines.



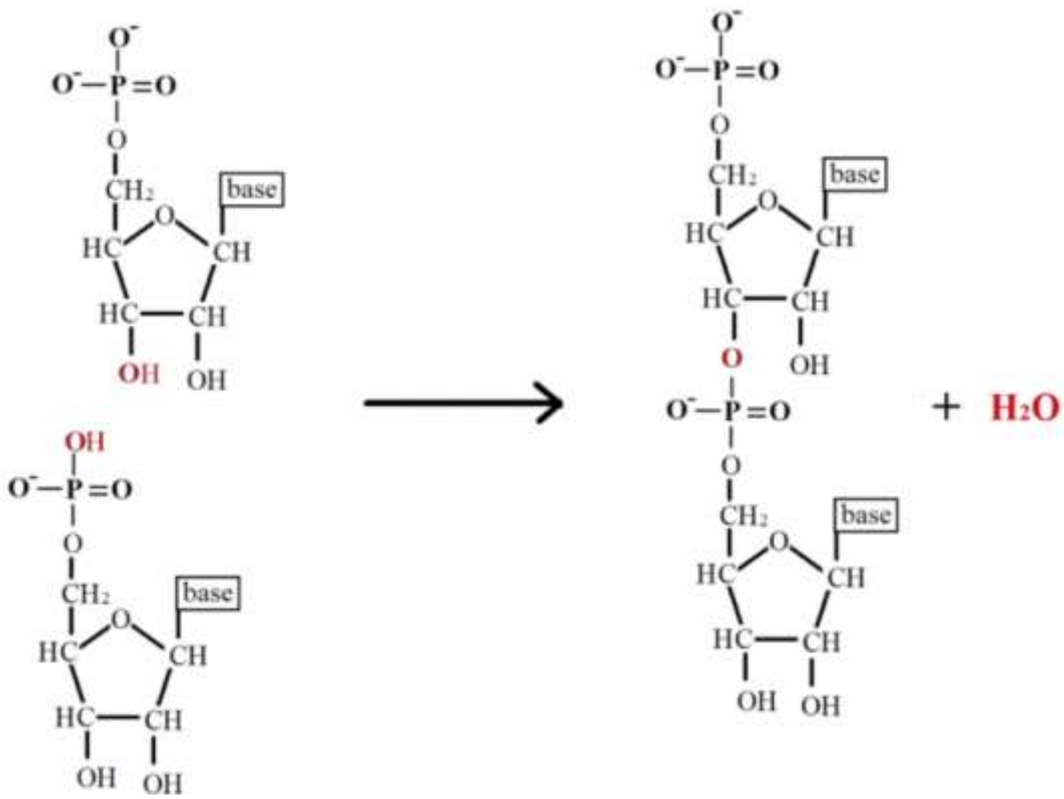
- **Pyrimidine bases** are of three types i.e., cytosine (C), thymine (T) and uracil (U). Thymine is only found in DNA while the uracil is only found in RNA.
- **Purine bases** are also two types i.e., adenine (A) and guanine (G).

Formation of nucleotide

- During the formation of a typical nucleotide, firstly nitrogenous base is linked with 1' carbon of pentose sugar. Such combination is called **nucleoside.**
- The bond that combines the base with sugar is called glycosidic bond (formed between C'1 of pentose sugar and N9 of purines or N1 of pyrimidines)
- When a phosphoric acid is linked with 5' carbon of pentose sugar of a nucleoside, the **nucleotide** is formed.
- A nucleotide with one phosphoric acid is called nucleoside monophosphate, with two phosphoric acids is called nucleoside diphosphate and with three phosphoric acids is called nucleoside triphosphate.
- The nucleotides which take part in the formation of DNA or RNA polymer must contain three phosphates but during their incorporation into DNA or RNA polymer each nucleotide losses its two terminal phosphates.

Nitrogenous Base	RNA (Ribo-nucleosides)	RNA (Ribo-nucleotides)	DNA (Deoxy-ribonucleosides)	DNA (Deoxy-ribonucleotides)
Adenine	Adenosine	AMP, ADP, ATP	d-Adenosine	dAMP, dADP, dATP
Guanine	Guanosine	GMP, GDP, GTP	d-Guanosine	dGMP, dGDP, dGTP
Cytosine	Cytidine	CMP, CDP, CTP	d-Cytidine	dCMP, dCDP, dCTP
Uracil/Thymine	Uridine	UMP, UDP, UTP	d-Thymidine	dTMP, dTDP, dTTP

Phosphodiester linkages



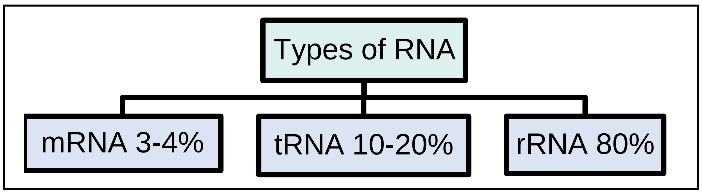
Formation of a phosphodiester bond

- The bond formed between phosphoric acids (H_3PO_4) and hydroxyl (OH) groups of pentose sugar is called ester linkage.

- In a polynucleotide chain one phosphoric acid is attached to the OH group of carbon no. 3 of pentose sugar in front while carbon No 5 of another pentose sugar behind it forming **phosphodiester linkage(C-O-P-O-C)** between consecutive nucleotides**.**
- When one nucleotide joins to other in formation of polynucleotide, the molecule that is released is not water but pyrophosphate (two phosphate groups bound together). When pyrophosphate is cleaved by the addition of water, a great deal of free energy is released.
- In this way nucleotides begin to link by phosphodiester bonds and a polymer of nucleotides (polynucleotide) is formed.

Ribonucleic acid (RNA)

- RNA is a polymer of ribonucleotides.
- Unlike DNA, RNA molecule is generally **single stranded** and does not form a double helix except **Reo virus**.
- But some RNA molecules have regions in which hydrogen bonds between A = U and G = C bases are formed between different regions of the same molecule thus coiled itself look like double stranded hair-pin loops.
- RNA is mostly present in cytoplasm but synthesized within nucleus by using one strand of DNA as templane through **transcription.**
- **RNA polymerase enzyme** is used in transcription. In eukaryotes, there are three types of RNA polymerases.
  1. RNA polymerase I transcribes rRNA
  2. RNA polymerase II transcribes mRNA
  3. RNA polymerase III transcribes tRNA.
- About 97% of the transcriptional output is non-protein coding in eukaryotes. So, they are called **non-coding RNA.**



Types of RNA

- There are **three** main types of RNA.

Messenger RNA (mRNA)

- Messenger RNA carries the genetic information from DNA in nucleus to ribosomes in the cytoplasm, where amino acids are arranged according to the information in mRNA to form specific protein molecule. This process is called **translation.**
- This type of RNA consists of a single strand of variable length.
- Its length depends upon the size of the gene as well as the protein for which it is taking the message.
- For example, for a protein molecule of 1,000 amino acids, mRNA will have the length of 3,000 nucleotides.
- Because every three nucleotides in mRNA encode a specific amino acid, such triplets of nucleotides along the length of mRNA are called **codons** of genetic codes.
- mRNA is about **3 to 4%** (BTB=3-5%) of the total RNA in the cell.

Ribosomal RNA (rRNA)

- It is the major portion of RNA in the cell, and may be up to **80%** of the total RNA.
- It is strongly associated with the ribosomal protein where **40 to 50%** of it is present.
- It is synthesized by the genes on DNA of several chromosomes found within the region of nucleus called nuclear organizer.
- On the surface of the ribosome the mRNA and tRNA molecules interact to translate the information from genes into a specific protein.
- rRNA acts as a **machinery** for the synthesis of proteins.
- rRNA is the catalytic component of the ribosome.
- rRNA has the largest size among RNA.
- The base sequence of rRNA of all organisms is simar therefore, there is only one type of rRNA.

Transfer RNA (tRNA)

- It comprises about **10 to 20%** (BTB = 15%) of the cellular RNA.
- Transfer RNA molecules are smallest among RNA, each with a chain length of **75 to 90** (BTB = 80) nucleotides.

- A tRNA is a single stranded molecule but it shows a duplex appearance at its some regions where complementary bases are bonded to one another.
- It shows a flat cloverleaf shape in two dimensional views.
- Its **5' end** always terminates in Guanine base while the **3' end** is always terminated with base sequence of CCA.(**mnemonic:C**ome **CA**t = **CCA**)
- Amino acid is attached to tRNA at 3 end.
- It transfers amino acid molecules to the ribosomes where peptide chains are being synthesized to form proteins.
- There is one specific tRNA for cach amino acid. So, the cell will have at least 20 kinds of tRNA molecules.
- **Sixty** tRNA have been identified. However, human cells contain about **45** different kinds of tRNA.
- **tRNA has three loops**:

1. Middle loop

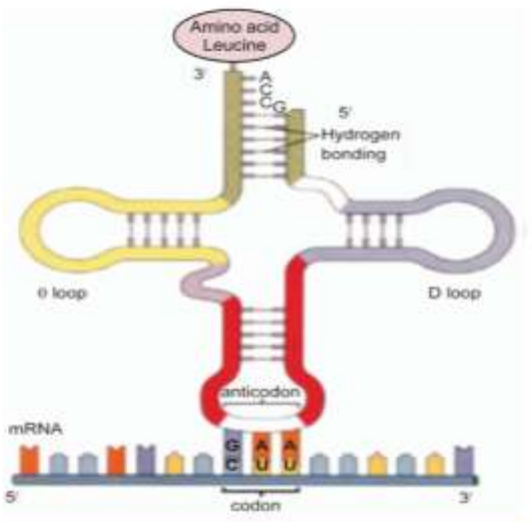
The middle loop in all the tRNA is composed of 7 bases, the middle three of which form the anticodon; it is complementary to specific codon of mRNA.

2. D-loop

D loop recognizes the activation enzyme.

3. Theta loop

Theta loop recognizes the specific place on the ribosome for binding during protein synthesis.



Cloverleaf Model of tRNA.

Mnemonic: MA = Middle **A**nticodon, **AD** = **A**ctivation of enzyme **D**-loop, **TB** = **T**heeta **B**inds ribosomes

Conjugated molecules

- **Two different molecules**, belonging to different categories, usually combine together to form conjugated molecules.

Glycoproteins (Carbohydrate + Protein)

- Carbohydrates combine covalently with proteins to form glycoprotein.
- Most of the cellullar secretions are glycoprotein in nature.
  (**mnemonic: SP** = **S**ecretion **P**roteins (glyco), not glycolipids)
- They function as hormones, transport proteins, structured proteins and receptors.
- The blood group antigens contain glycoproteins, which also play an important role in blood grouping.
- They are also present in egg albumin.
- Glycoproteins are integral components of plasma membrane.
- Glycoprotein and glycolipids have a structural role in the extracellular matrix of animals and bacterial cell wall.

Glycolipids (Carbohydrate + Lipid)

- These are lipids attached with a carbohydrate through glycosidic bond.
- Glycolipids are **complex lipids** containing one or more simple sugars in connection with long fatty acids or alcohol.
- Glycolipids are present in white matter of brain and myelin sheath of nerve fibres and chloroplast membrane.
- They are also integral structural components of plasma membrane.

Lipoproteins (Lipid+ Protein)

- Lipoprotein formed by combination of lipids and proteins.
- It is **basic structural framework** of all types of membranes in the cells.
- They occur in milk, blood, cell nucleus, egg yolk membrane and chloroplasts of plants.

Nucleoproteins (Nucleic acid + Protein)

- Nucleic acids have special affinity for **basic** proteins. They are combined together to form nucleoproteins.
- They are present in ribosomes.
- The **nucleohistones** (histone protein+nucleic acid) are present in chromosomes.
- They play an important role in regulation of gene expression.
- These are slightly acidic and soluble in water.

Shortlisting

- Approximately **25 elements** bio elements. Humans have **16** of them. Out of which **6** are major.
- These may be organic or inorganic.
- **Proteins** are the **most abundant** organic compounds while nucleic acids are the most essential
- about 20 percent water in bone cells and 85 percent (FTB = 85 to 90%) in brain cells.
- Hydrogen bonding is responsible for exceptional thermodynamic properties of water.
- The specific heat of vaporization of water is **574 kcal/kg.**
- The specific heat capacity of water is **4.184 J/g°C or 1 cal/g°C**.
- carbohydrates are defined as "**polyhydroxy aldehydes or ketones,** or **complex substances** which on hydrolysis yield polyhydroxy aldehyde or ketone subunits."
- They are classified into monosaccharides (having one monomer), oligosaccharides (having 3 to 7 monomers), polysaccharides (having 1000s of units).
- **Enantiomers**: These are mirror images of each other, differing in configuration at every chiral center (e.g., D-glucose and L-glucose).
- **Diastereoisomers**: Non-mirror image isomers differing in configuration at one or more (but not all) chiral centers (e.g., D-glucose and D-mannose).
- **Epimers**: A subtype of diastereoisomers differing in configuration at only one specific chiral center (e.g., D-glucose and D-galactose).
- **Anomers**: Isomers that differ at the newly formed chiral center after ring closure, typically at the C-1 carbon (e.g., α-D-glucose and β-D-glucose).
- Our blood normally contains 0.08% glucose.
- For the synthesis of **10g** of glucose **717.6 kcal** of solar energy is used.
- **Sucrose**: Glucose + Fructose, α-1,2 glyosidic linkage
- **Lactose**: Galactose + Glucose, β-1,4 glyosidic linkage
- **Maltose**: Glucose + Glucose, α-1,4 glyosidic linkage

<u>**Polysaccharides**</u>

- **Starch**: Glucose, α-1,4 and α-1,6 glyosidic linkages
- **Cellulose**: Glucose, β-1,4 glyosidic linkage
- **Glycogen**: Glucose, α-1,4 and α-1,6 glyosidic linkages
- **Chitin**: N-acetylglucosamine, β-1,4 glyosidic linkage
- About **170** types of amino acids have been found to occur in cells and tissues.
- Most of the proteins are however, made of **20** types of amino acids.
- **Insulin** is composed of 51 amino acids(49 peptide linkages) in two chains. One of the chains had 21 amino acids and the other had 30 amino acids
- Similarly, **Haemoglobin** is composed of 574 amino acids (570 peptide linkages) in four chains, two alpha and two beta chains. Each alpha chain contains 141 amino acids, while each beta chain contains 146 amino acids
- **α-helix 3.6** amino acids in each turn.
- **Primary structure**: Comprises the sequence of amino acids linked by peptide bonds.
- **Secondary structure**: Includes α-helices and β-sheets formed through hydrogen bonds.
- **Tertiary structure**: Refers to three-dimensional folding driven by side chain interactions, involving disulfide bonds, ionic bonds, hydrogen bonds, and hydrophobic interactions.
- **Quaternary structure**: Represents the assembly of multiple polypeptide chains, connected by the same types of bonds as found in the tertiary structure — disulfide, ionic, hydrogen, and hydrophobic interactions.
- **Sickle cell Hb** shows only one difference from normal Hb i.e., glutamic acid is replaced by valine at position number six in both β-chains.
- The most **abundant** protein:
  1. Earth = rubisco
  2. Human body = collagen type (I)
  3. Blood = albumin
- Lipids classes:
- **Acylglycerols**: Glycerol + fatty acids; triacylglycerols are most abundant.
- **Waxes**: Long-chain esters; melting point is high, e.g., beeswax. $C_{15}H_{31}COOC_{30}H_{16}$ (STB)
- **Phospholipids**: Amphiphilic lipids; form cell membranes.
- **Glycolipids**: Lipid + carbohydrate; present in membranes.
- **Terpenoids**: Isoprene polymers; C_5H_8 units, e.g., vitamin A.
- **Steroids**: Four-ring structure; includes cholesterol, hormones. 17 carbon nucleus

- **Prostaglandins**: Arachidonic acid derivatives; 20-carbon structure.
- Nucleic acids were first isolated (in 1869) by a Swiss physician, Fredrick miescher from the nuclei of **pus cells** (white blood cells) and **sperm** of salmon fish.
- **Nucleotide**: Consists of a phosphate group, a sugar, and a nitrogenous base.
- **Nucleoside**: Contains only a sugar and a nitrogenous base, without the phosphate group.
- In polynucleotides there is **phosphodiester linkage(C-O-P-O-C)** between consecutive nucleotides**.**
- **Glycolipids**: Lipid + carbohydrate; function in cell recognition and signaling.
- **Lipoproteins**: Lipid + protein; transport lipids like cholesterol and triglycerides in blood.
- **Glycoproteins**: Protein + carbohydrate; involved in immune response and intercellular communication.
- **Nucleoproteins**: Protein + nucleic acid; key roles in genetic material organization (e.g., chromatin) and protein synthesis (e.g., ribosomes).

Past MDCAT paper + Most repeated MCQs

Introduction to biological molecules and importance of water

1. Of the total weight of a bacterial cell, carbohydrates constitute only:
A. 1% B. 2% C. 3% D. 4%

2. 18% of the total weight of a mammalian cell is the:
A. Water B. Proteins C. Carbohydrates D. Lipids

3. The amount of heat required to raise the temperature of 01g of water by 01 degree:
A. Specific Heat Capacity C. Electronegativity
B. Heat of vaporization D. Electron affinity

4. Name the process that involves the breakdown of large molecules into smaller ones utilizing water molecules:
A. Hydrolysis B. Reduction C. Oxidation D. Condensation

5. All of the following are formed through condensation reactions except:
A. Glucose B. Amylopectin C. Insulin D. Triglyceride

6. Ester bond is present in all except:
A. Two nucleotides C. Glycerol and fatty acid
B. Glycerol and phosphoric acid D. Two amino acids

7. All are uses of water by living organisms except:
A. as lubricant & for protection C. used in metabolism
B. as cooling agent & thermo-stabilizer D. source of energy

8. Value of heat capacity of water is:
A. 1.0 cal/g B. 10 cal/g C. 100 kcal/kg D. 574 kcal/g

9. What will happen if water content of protoplasm is reduced as low as 10 percent?
A. Cell will die C. No effect
B. Metabolism will pace up D. Metabolism will slow down

10. Temperature of water rises and falls more slowly as compared to other liquids due to its ______ property.
A. Polar nature B. Heat of vaporization C. Specific Heat D. Ionization

Carbohydrates

11. A biochemical test used for detection of reducing sugars is:
A. Biuret test B. Benedict test C. Spot test D. Iodine test

12. All of the following elements are found in all carbohydrates except:
A. Carbon B. Hydrogen C. Nitrogen D. Oxygen

13. Ratio of hydrogen and oxygen in carbohydrates is:
A. 1:1 B. 2:2 C. 1:2 D. 2:1

14. General formula of carbohydrates is:
A. $Cn(H_2O)_n$ B. $(CH_2O)_n$ C. $Cx(H_2O)y$ D. $Cn(H_2O)_{n-1}$

15. These are primary products of photosynthesis:
A. Carbohydrates B. Proteins C. Lipids D. Nucleic acids

16. It is not true for carbohydrates:
A. Carbohydrates are used in synthesis of lipids
B. Carbohydrates are less soluble than fats in water
C. Carbohydrates form a component of nucleic acids

D. Carbohydrates are utilized in the synthesis of amino acids

17. It is not a carbohydrate:
A. Starch B. Glycogen C. Chitin D. Cutin

18. Most of the energy from glucose is released by breakdown of:
A. C—C B. C—N C. C—H D. C—OH

19. $Cn(H_2O)_n$ is not applicable on:
A. Glucose B. Fructose C. Deoxyribose D. Ribose

20. The reducing sugars are so called because they can ____ electron/s:
A. Donate B. Gain C. Share D. Excite

21. All of the following are polymers of glucose except:
A. Glycogen B. Cellulose C. Amylase D. Amylopectin

22. Glycosidic bond cannot be found in:
A. Glucose B. Cellulose C. Glycogen D. Lactose

23. The covalent bond between two monosaccharides is called:
A. Peptide Linkage B. Glucosidic linkage C. Glycosidic linkage D. Ester linkage

24. Most familiar disaccharide is:
A. Maltose B. Lactose C. Sucrose D. Cellobiose

25. An important sugar occurring only in animals is:
A. Glucose B. Lactose C. Sucrose D. Fructose

26. Hydrolysis of which of the following would yield only glucose?
A. Lactose B. Maltose C. Cellulose D. Cellulase

27. Structural polysaccharides include:
A. Cellulose, Hemicellulose, Chitin C. Cellulose, Starch, Glycogen
B. Cellulose, Starch, Chitin D. Cellulose, Glycogen, Chitin

28. It is an example of polysaccharide that is soluble in hot water:
A. Amylose B. Glycogen C. Cellulose D. Amylopectin

29. Most abundant carbohydrate in nature is:
A. Starch B. Glycogen C. Hemicellulose D. Cellulose

30. Type of glycosidic linkage present in cellulose is:
A. α-1, 4 B. β-1, 4 C. α-1, 4 and α-1, 6 D. β-1, 4 and β-1, 6

Proteins

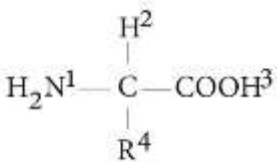
31. Primary structure of a protein molecule does not comprise:
A. Number of amino acids C. Size of protein molecule
B. Sequence of amino acids D. Shape of protein molecule

32. All proteins in living organism always show following structural levels:
A. Primary & secondary structure C. Tertiary & quaternary structure
B. Secondary & tertiary structure D. Secondary & quaternary structure

33. In quaternary structure, polypeptide chains are aggregated and held together by all of the following except:
A. Hydrophobic interactions B. Glycosidic linkages C. Hydrogen bond D. Ionic bond

34. It is an example of globular protein:
A. Myosin B. Fibrin C. Fibrinogen D. Keratin

35. An amino acid molecule has the following structure... Which two of the groups combine to form a peptide link?



A. 1 and 2 B. 1 and 3 C. 2 and 3 D. 3 and 4

36. The sequence of amino acids in a protein is determined by sequence of ____ in ____:
A. Bases, tRNA B. Nucleotides, tRNA C. Nucleotides, DNA D. Nucleotides, mRNA

37. Bond that is most sensitive to rise in temperature is:
A. Peptide bond B. Ionic bond C. Disulphide bond D. Hydrogen bond

38. Number of carbon atoms in simplest amino acid are:
A. 2 B. 3 C. 4 D. 5

39. Which amino acid is essential for formation of disulphide linkages in proteins?
A. Glycine B. Alanine C. Serine D. Cysteine

40. β-Pleated sheet is an example of:

A. Primary structure B. Secondary structure C. Tertiary structure D. Quaternary structure

Lipids

41. Chemical compounds that are defined on base of physical properties are:
A. Carbohydrates B. Proteins C. Lipids D. Nucleic acids
42. Lipids store energy due to higher proportion of:
A. C—C bonds B. C—O bonds C. C—H bonds D. C—N bonds
43. This is an example of conjugated molecule of lipid:
A. Acylglycerol B. Phospholipid C. Glycoprotein D. Glycolipid
44. Most important components of triglycerides are:
A. Glycerols B. Fatty acids C. Ketones D. Isoprenoid
45. Functional group of fatty acid is:
A. $-CH_3$ B. –OH C. –CHO D. –COOH
46. Which characteristics are applicable to all fats and oils?

Fats	Oils
I) Saturated fat acids	Unsaturated fat acids
II) Solid at room temperature	Liquid at room temperature
III) Can be crystalized	Cannot be crystallized
IV) Obtained from animals	Obtained from plants

A. I & II B. I & III C. I, II & III D. I, II, III & IV
47. Saturated fatty acids have C-C double bonds:
A. 0 B. 2 C. 1 D. 3
48. All of the following nitrogenous bases are found in phospholipids except:
A. Choline B. Cytosine C. Ethanolamine D. Serine
49. Fatty acids are found in all of the following except:
A. Acylglycerols B. Phospholipids C. Terpenoids D. Waxes
50. Formation through condensation of which of the following is not associated with release of water:
A. Amylose B. mRNA C. Fibrin D. Carotenoid

Nucleic acids and conjugated molecules

51. Identify the monocyclic structure:
A. Uracil B. Guanine C. Adenine D. None
52. Which of the following represents high energy bonds in ATP?
A. Ribose – Adenine
B. Ribose – Phosphate
C. Phosphate – Adenine
D. Phosphate – Phosphate

53. In a typical nucleotide the nitrogenous base is attached to ______ carbon of pentose sugar:
A. C-1 B. C-5 C. C-3 D. C-6
54. Phosphodiester bond is formed between:
A. Two phosphate groups
B. Two phosphates & one hydroxyl group
C. One phosphate & two hydroxyl groups
D. Two phosphate & two hydroxyl groups
55. All of the following are examples of dinucleotides except:
A. ADP B. NAD C. NADP D. FAD
56. It is the major proportion of RNA in the cell:
A. mRNA B. tRNA C. rRNA D. rDNA
57. Messenger RNA carries the genetic information from:
A. Nucleus to nucleolus B. Nucleolus to ribosome C. DNA to tRNA D. DNA to ribosome
58. Which one is correct about the URACIL?
A. It is a nucleotide
B. It contains purine nitrogen
C. It is used to form DNA
D. It is used to form RNA
59. Conjugated molecules are of ______ significance:
A. Structural significance only
B. Functional significance only
C. Structural and functional significance
D. Has no structural and functional role
60. Which of the following conjugated molecule is incorrectly matched?
A. Lipo-proteins ------- cell membrane
B. Glyco-proteins ------- cell surface antigens/ receptors
C. Glyco-lipids ------- cell wall
D. Nucleic acids ------- chromosomes

Answers and Explanations

Q	Ans	Explanation
Q.1	C	Carbohydrate is 3% of the total Bacterial cell.
Q.2	B	Protein is 18% of the total mammalian cell.
Q.3	A	Heat capacity is the amount of energy to increase the temperature of that substance by 1°C.
Q.4	A	Sucrose + H_2O → Glucose + Fructose
Q.5	A	Larger organic macromolecules are formed through condensation reactions. Glucose is formed through photosynthesis.
Q.6	D	Two amino acids are connected through peptide bond.
Q.7	D	Mostly living organisms use glucose as major energy source.
Q.8	A	To increase 1°C temperature of 1 gram of water we need 1 cal.
Q.9	A	A 70–90% of body weight consists of water. Below 10% water in protoplasm, the survival of cell becomes impossible.
Q.10	C	Water has ability to absorb a lot of heat with a little change in its own temperature.
Q.11	B	Biuret test is to detect proteins. Spot test is for fats. Iodine test is for polysaccharides.
Q.12	C	C, H and O are essential elements found in all carbohydrates while N is non-essential.
Q.13	D	They have same ratio as in H_2O (2:1).
Q.14	C	$C_n(H_2O)_n$ and $(CH_2)_n$ are general formulae of monosaccharides. $C_n(H_2O)_{n-1}$ is disaccharide.
Q.15	A	All other organic compounds are synthesized from carbohydrates.
Q.16	B	Fats are not soluble in water. They are hydrophobic compounds.
Q.17	D	Cutin is part of the waxy plant cuticle.
Q.18	C	C—H linkage is principal source of energy.
Q.19	C	Formula of deoxyribose is $C_5H_{10}O_4$.
Q.20	A	Reducing sugars act as reducing agents and release electrons.
Q.21	C	Amylase is enzyme and polymer of amino acids.
Q.22	A	Monosaccharides being single sugars do not have glycosidic bond.
Q.23	C	Glycosidic linkage is found between sugar units. Peptide bond in amino acids in proteins. Ester linkage in lipids and nucleic acids.
Q.24	C	Sucrose is most common disaccharide.
Q.25	B	Lactose is a milk sugar.
Q.26	B	Lactose → Galactose + Glucose Sucrose → Fructose + Glucose Cellulase → Amino acids
Q.27	A	Starch and glycogen are storage polysaccharides.
Q.28	A	Amylose is soluble in hot water.
Q.29	D	Cellulose is most abundant carbohydrate.
Q.30	B	Cellulose is unbranched polymer of β-glucose.
Q.31	C	Primary structure is sequence-based, not size-based.
Q.32	A	Primary and secondary found in all proteins.
Q.33	B	Glycosidic linkages are for carbohydrates, not proteins.
Q.34	C	Fibroin is silk protein. Fibrinogen is globular while fibrin is fibrous.
Q.35	B	Peptide bond is formed between 1) NH_2 and 3) COOH group.
Q.36	D	DNA determines primary sequence of proteins.
Q.37	D	Ionic bond is most sensitive to pH change.
Q.38	A	Glycine has 2 carbon atoms.
Q.39	D	Cysteine is sulfur-containing amino acid.
Q.40	B	β-pleated sheet is secondary structure.
Q.41	C	Lipids are defined on solubility.
Q.42	C	C–H bond are potential source of energy and lipids have them in double the amount as compared to carbohydrates.
Q.43	D	Glycolipids are derivatives; others are simple.
Q.44	B	Depends on fatty acid type.
Q.45	D	–COOH is carboxylic group.

<table>
<tr><td>Q.46</td><td>D</td><td>
<table>
<tr><th>Saturated fatty acid</th><th>Unsaturated fatty acid</th></tr>
<tr><td>No double bonds between carbon atoms</td><td>Upto six double bonds</td></tr>
<tr><td>Straight chain</td><td>Ringed / Branched</td></tr>
<tr><td>Solid at room temperature</td><td>Liquid at room temperature</td></tr>
<tr><td>Fats</td><td>Oils</td></tr>
<tr><td>Animals</td><td>Plants</td></tr>
<tr><td></td><td>More useful for living things.</td></tr>
</table>
</td></tr>
<tr><td>Q.47</td><td>A</td><td>No double bond = saturated.</td></tr>
<tr><td>Q.48</td><td>B</td><td>Cytosine is nucleic acid base.</td></tr>
<tr><td>Q.49</td><td>C</td><td>Terpenoids have isoprenoid not fatty acids.</td></tr>
<tr><td>Q.50</td><td>D</td><td>Carotenoids form without water release.</td></tr>
<tr><td>Q.51</td><td>A</td><td>Adenine and guanine are bi-cyclic. Uracil is monocyclic.</td></tr>
<tr><td>Q.52</td><td>D</td><td>Phosphodiester bond is a high energy bond.</td></tr>
<tr><td>Q.53</td><td>A</td><td>Nitrogenous base is attached to C1.</td></tr>
<tr><td>Q.54</td><td>C</td><td>Phosphodiester bond is formed between one phosphate and two hydroxyl groups.</td></tr>
<tr><td>Q.55</td><td>A</td><td>ADP is a mononucleotide, not a dinucleotide.</td></tr>
<tr><td>Q.56</td><td>C</td><td>rRNA constitutes about 80% of total RNA found in any cell.</td></tr>
<tr><td>Q.57</td><td>D</td><td>mRNA is formed from DNA through transcription and translated at ribosomes.</td></tr>
<tr><td>Q.58</td><td>D</td><td>• Uracil is a pyrimidine nitrogenous base
• Used to form RNA as it contains ribose sugar.</td></tr>
<tr><td>Q.59</td><td>C</td><td>Conjugated molecules have both structural (plasma membrane) and functional (regulation of gene expression) significance.</td></tr>
<tr><td>Q.60</td><td>C</td><td>Cell wall composition
• Plants: Cellulose, Fungi: Chitin, Bacteria: peptidoglycan.</td></tr>
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